11 Physics

Wave Properties

THE NATURE OF WAVES

Waves have the following characteristics.

To know

1. They consist of disturbances progressing through all or part of a medium.
2. An energy change or source of energy is needed to set up the waves.
3. ***Energy*** is transferred as the wave progresses through the medium.
4. The medium itself is not moved from one point to another.

Example

Energy is moving "under" the boat but the boat responds by moving up + down.

On the ocean, a boat bobs up and down as the waves move underneath it. The boat does not move with the wave - energy is being transmitted but the medium does not move with it.

The water particles merely oscillate up and down about some equilibrium position in accordance to the energy as it reaches them.

Mechanical waves (e.g. water waves, sound) need a medium to travel in, while ***electromagnetic waves*** (e.g. light) do not need a medium.

MECHANICAL WAVES

Water waves at the beach are caused by the wind disturbing the surface.

This type of wave needs an ***energy source to produce a disturbance*** and an ***elastic medium to transmit the disturbance***.

An elastic medium has particles that occupy a mean position from which they can be temporarily displaced by an outside force. They return to their original position after the force has ceased.

e.g. Air particles behave elastically.

There are two types of mechanical waves - ***longitudinal*** and ***transverse***.

Longitudinal Wave (e.g. sound)

- The particles of the medium move ***parallel*** to the direction of propagation of the wave disturbance.

High pressure Low pressure

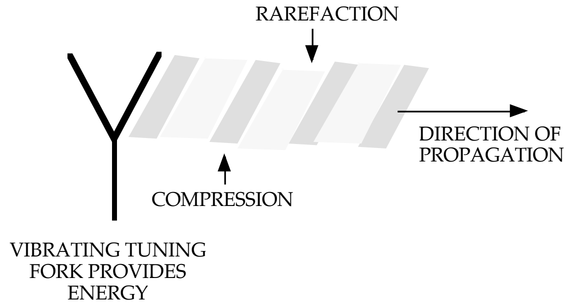
- Longitudinal waves are a series of ***compressions*** and ***rarefactions***.

- ***Compression*** - particles have been pushes closer together.

  i.e. ***High pressure area.***

- ***Rarefaction*** - particles are further apart.

  i.e. ***Low pressure area.***



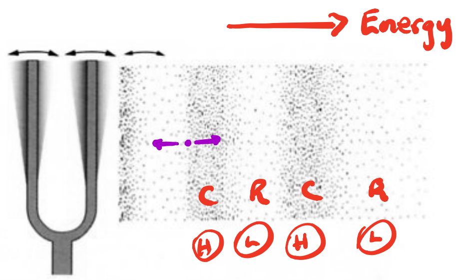
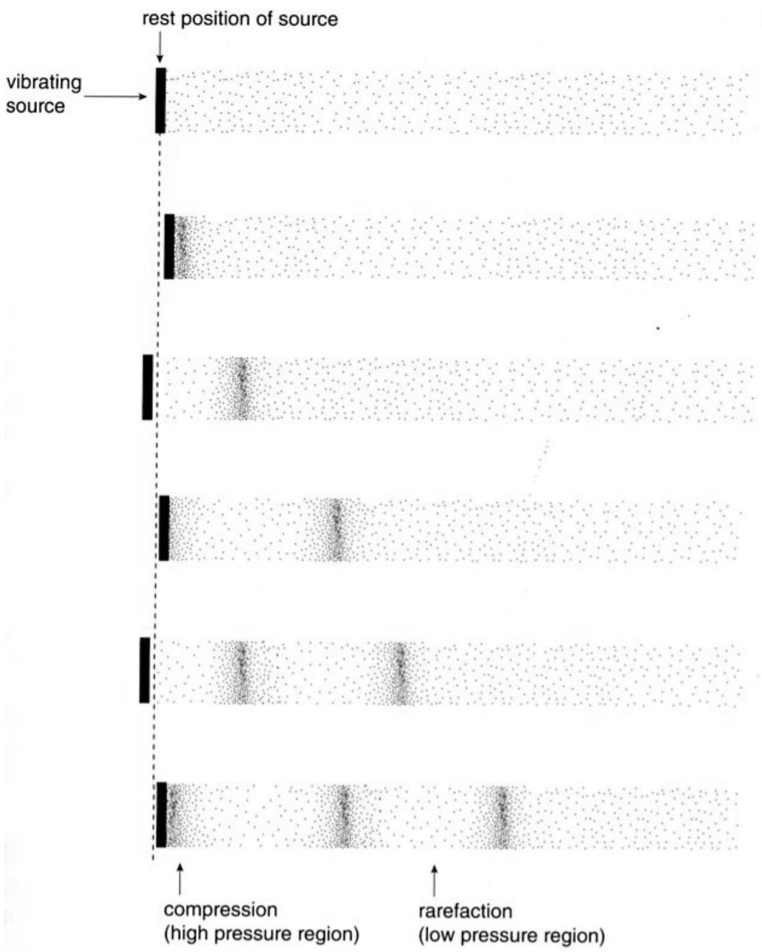


Figure 5.3
The rapid vibration of a tuning fork makes the air molecules around it vibrate in much the same way as the human larynx or a loudspeaker—creating the compressions and rarefactions of a pressure wave that we perceive as sound.

After the disturbance has passed a given point, the particles return to the equilibrium or mean position.

The distance a particle is displaced from its equilibrium position depends on the energy - the greater the energy, the greater the displacement.



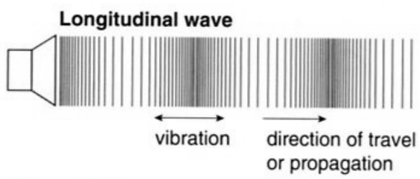


Figure 5.7
In a longitudinal wave, the vibrations of the individual particles are in the same direction as the direction of wave propagation.

Figure 5.8
In a medium in which there is no sound, the particles are evenly spread. As the source moves backwards and forwards, the continual vibration produces a series of compressions and rarefactions moving continuously away from the source. In this way, sound energy is propagated through a medium. Once the source ceases to vibrate, the particles in the medium return to approximately the same position. There is no net movement of the particles.

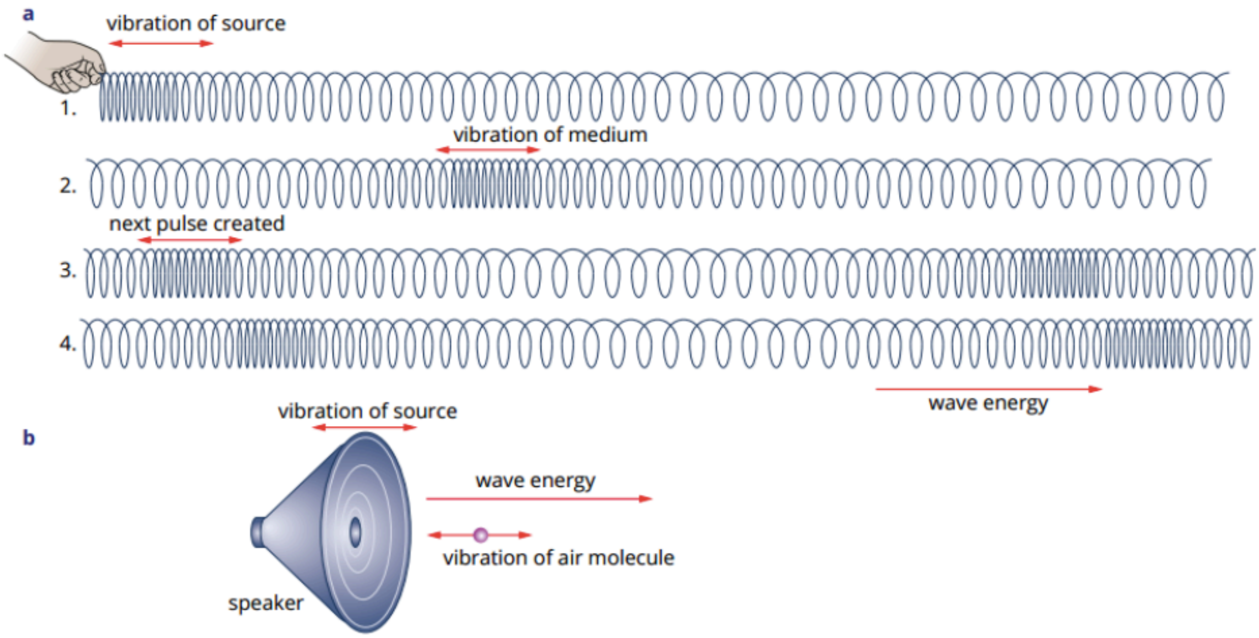
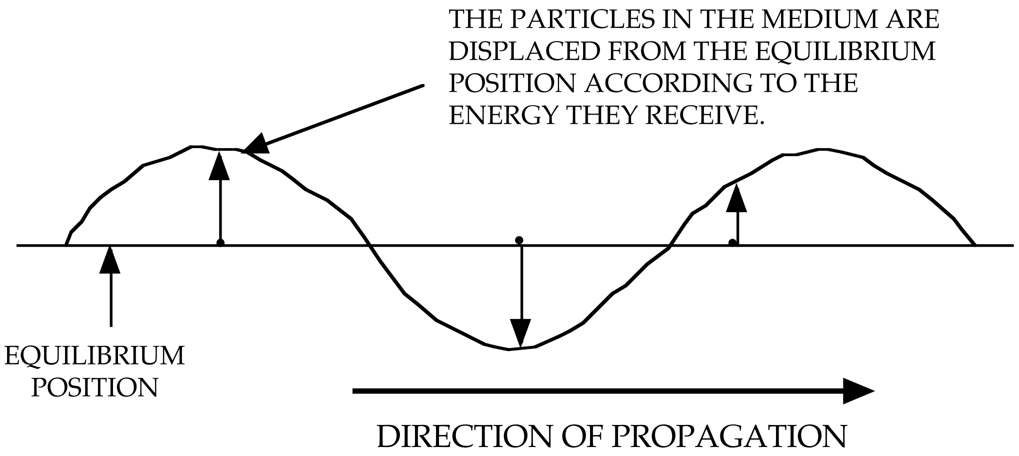


FIGURE 10.1.4 (a) When the direction of the vibrations of the medium and the direction of travel of the wave energy are parallel, a longitudinal wave is created. This can be demonstrated with a slinky. (b) Sound waves are longitudinal.

Transverse Waves (e.g. water waves)

- The particles of the medium move ***perpendicular*** to the direction of propagation of the wave disturbance.



- The particles vibrate about their equilibrium position in a periodic fashion.

  i.e. They move up and down only - they don't move with the wave front as it moves through the medium.

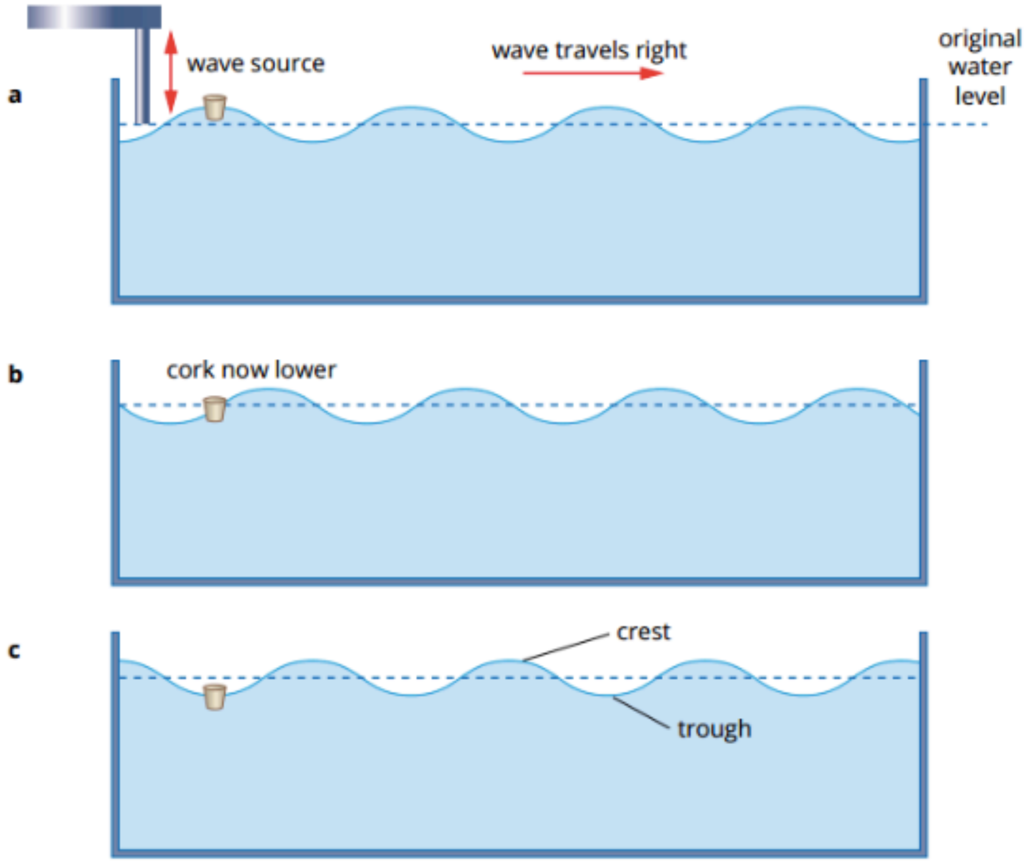


FIGURE 10.1.3 A continuous water wave moves to the right showing an example of a transverse wave. As it does so the up and down displacement of the particles transverse to the wave motion can be monitored using a cork. The cork simply moves up and down as the wave passes through it.

In both transverse and longitudinal waves, ***kinetic energy is transferred through the medium without the medium as a whole being displaced***.

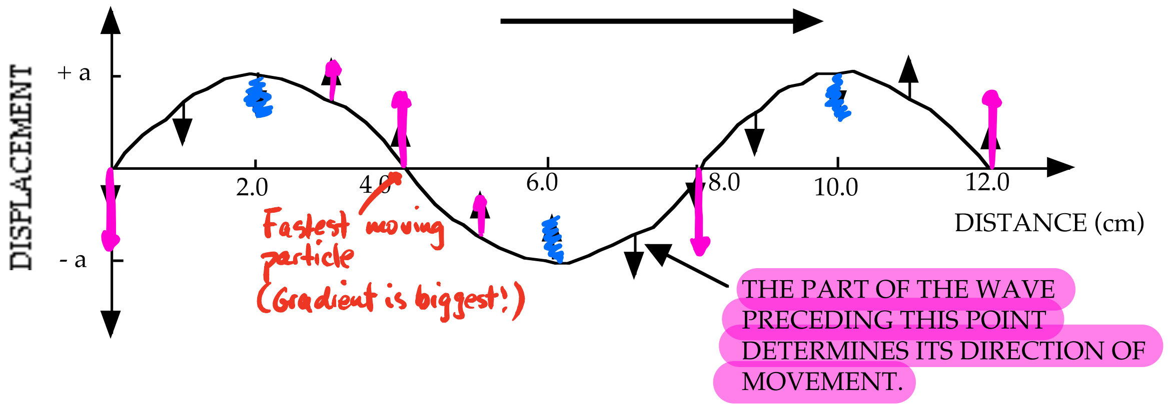
GRAPHING WAVE MOTION

There are two types of graphs that are drawn to represent a wave motion, namely ***displacement - distance*** and ***displacement - time*** graphs.

Displacement - Distance Graph

Like a photo of the whole medium

- This graph ***shows all of the particles of the medium at a given instant*** and their respective displacement from the equilibrium position.



- The arrows on the diagram above show the direction each particle is moving at that instant.

- The energy of the wave that is about to reach the point (i.e. the preceding part of the graph) determines whether the movement will be up or down.

- Even though longitudinal waves cause the particles of the medium to move backwards and forwards rather than up and down, this motion is still represented on a graph like the one above.

Displacement - distance graph.

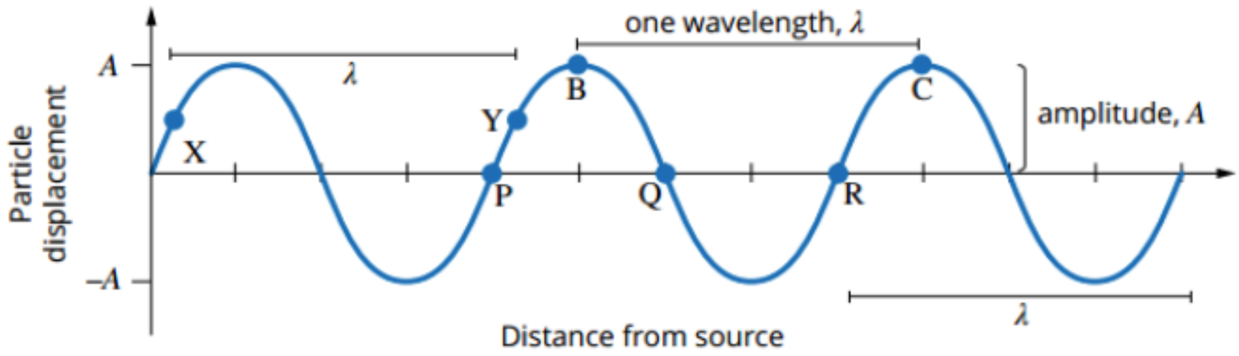
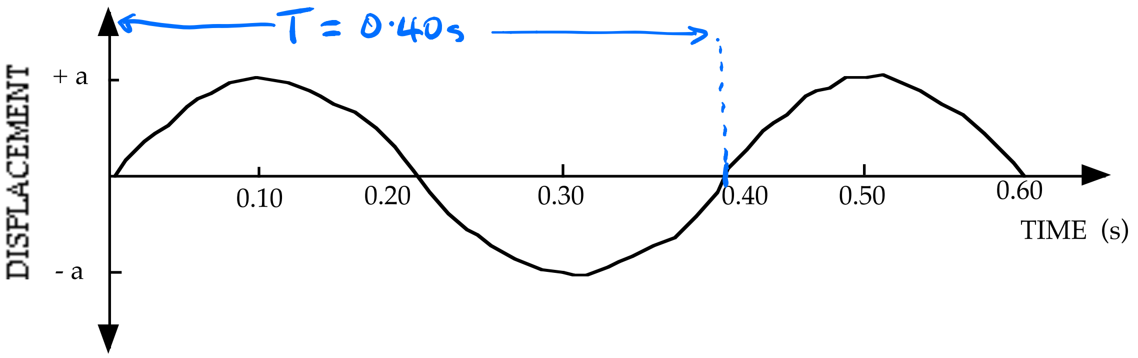


FIGURE 10.2.2 A sine wave representing the particle displacements along a wave over a distance.

Displacement - Time Graph

Like a movie of one point moving up/down or backwards/forwards.

- This graph represents ***one point in the medium*** and shows how it is moving with respect to time.



- This graph is used to ***represent both transverse and longitudinal wave movement***.

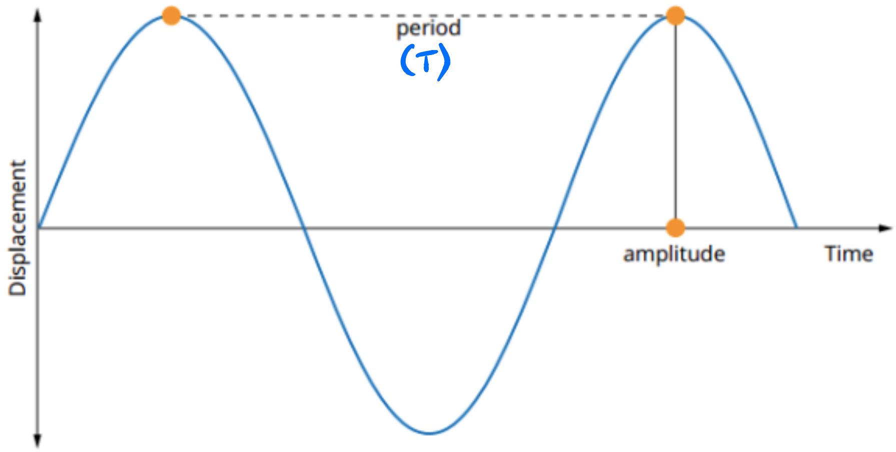


FIGURE 10.2.3 The graph of displacement versus time from the source of a transverse wave shows the movement of a *single point* on a wave over time as the wave passes through that point.

Read and learn!

CHARACTERISTICS OF WAVES

Phase

This is the ***position and motion of a particle at any instant***.

Particles having the ***same displacement from their equilibrium position*** and ***moving in the same direction*** are in phase.

Wavelength (λ)

This is the distance between two successive particles that are in phase.

Amplitude (a)

The maximum displacement of the particles from their equilibrium position.

Wave Speed (c) c = speed of light v = speed of .....

The distance travelled by the wave per unit time.

i.e. It's a velocity.

ν = nu?

Frequency (f,ν)

The number of waves that pass a given point per unit time.

This is the same as the frequency of vibration of the energy source.

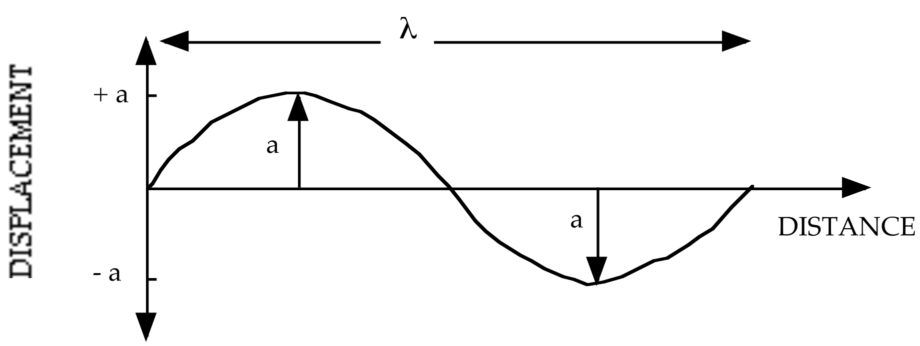
Period (T)

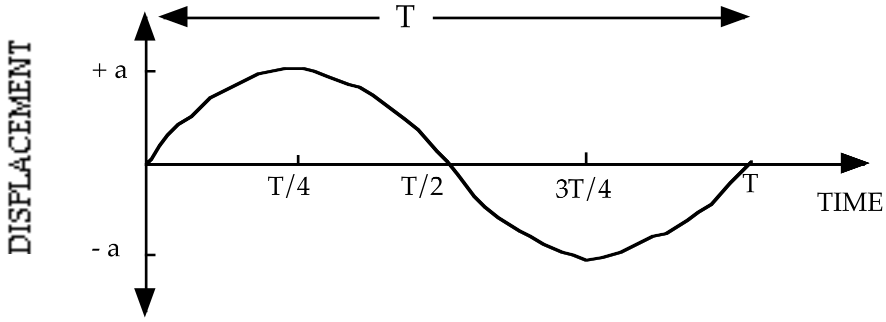
The time taken for one wavelength to pass a given point.

The time taken for a particle to undergo one complete oscillation.

$$\boxed{\mathbf{T = 1/f \quad \text{or} \quad T = 1/\nu}}$$

where f, ν = frequency (Hz)
T = period (s)





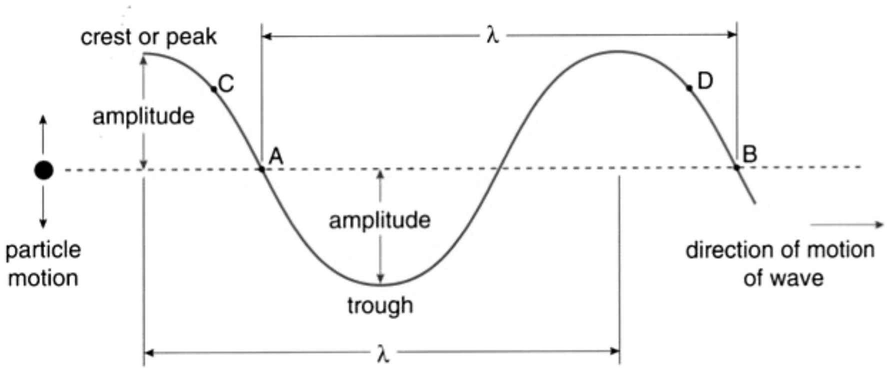


Figure 5.11
A simple pictorial representation of a transverse wave.

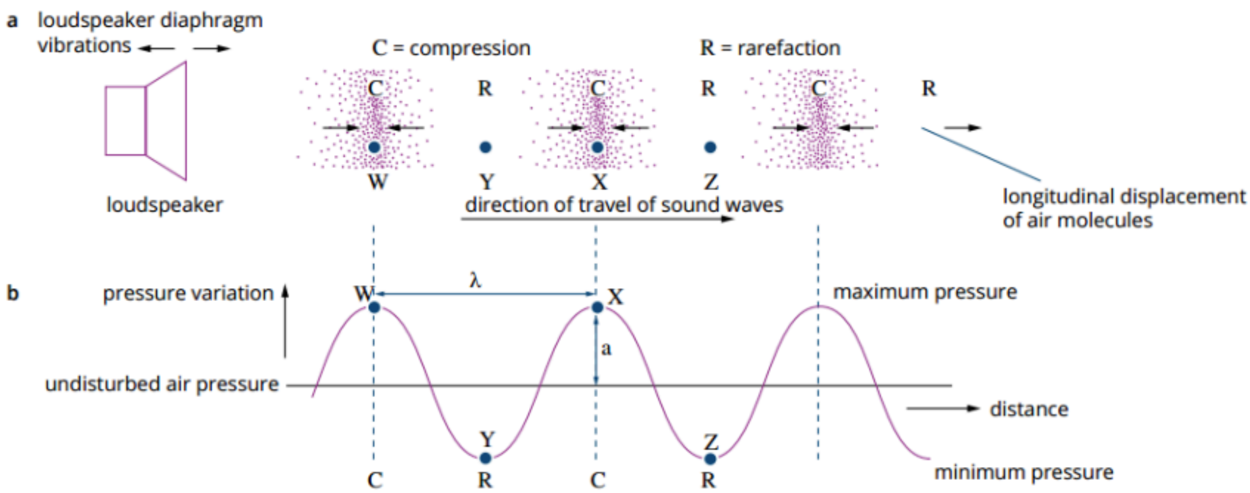


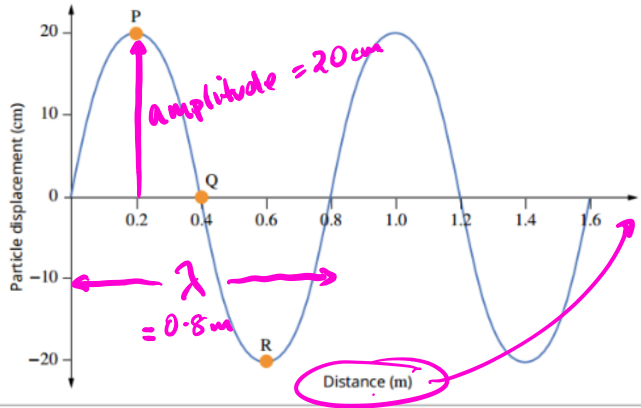
FIGURE 10.1.7 A longitudinal wave compared to a transverse wave. The compressions in a longitudinal wave are equivalent to a crest or peak in a transverse wave, and the rarefactions are equivalent to a trough.

Examples

1.

DISPLACEMENT–DISTANCE GRAPHS 1

The displacement–distance graph below shows a snapshot of a transverse wave as it travels along a spring towards the right. Use the graph to determine the amplitude and the wavelength of this wave.

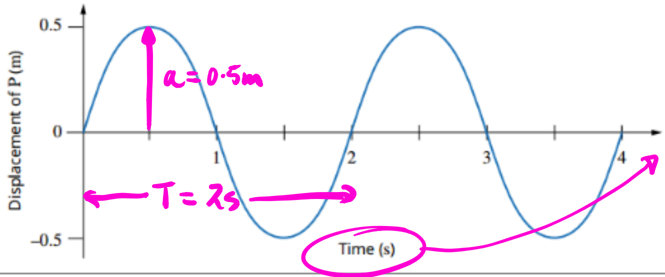


Thinking	Working
Amplitude on a displacement–distance graph is the distance from the average position (Q) to a crest (P) or a trough (R). Read the displacement of a crest or a trough from the vertical axis. Convert to SI units where necessary.	Amplitude = 20 cm = 0.2 m
Wavelength is the distance for one complete cycle. Any two consecutive points in phase and at the same position on the wave could be used.	The first cycle runs from the origin through P, Q, R to intersect the horizontal axis at 0.8 m. This intersection is the wavelength. Wavelength $\lambda = 0.8$ m

2.

DISPLACEMENT–TIME GRAPHS 2

The displacement–time graph below shows the motion of a single part of a rope (point P) as a wave passes by travelling to the right. Use the graph to find the amplitude, period and frequency of the wave.



Thinking	Working
The amplitude on a displacement–time graph is the displacement from the average position to a crest or trough. Note the displacement of successive crests and/or troughs on the wave and carefully note units on the vertical axis.	Maximum displacement is 0.5 m Therefore amplitude = 0.5 m
Period is the time it takes to complete one cycle and can be identified on a displacement–time graph as the time between two successive points that are in phase. Identify two points on the graph at the same position in the wave cycle, e.g. the origin and $t = 2$ s. Confirm by checking two other points, e.g. two crests or two troughs.	Period, $T = 2$ s
Frequency can be calculated using $f = \frac{1}{T}$, measured in hertz (Hz).	$f = \frac{1}{T} = \frac{1}{2} = 0.5$ The frequency is 0.5 Hz.

Wave Equation

The velocity of a wave is given by:

$$\boxed{\mathbf{c} = \mathbf{f\lambda} \qquad \text{or} \qquad \mathbf{c} = \mathbf{\nu\lambda}}$$

$v = f\lambda$
$c = f\lambda$ (for light)

where $\quad c \;=\;$ velocity of the wave (ms^{-1})
$\quad f, \nu \;=\;$ frequency of the wave (Hz)
$\quad \lambda \;=\;$ wavelength of the wave (m)

This equation applies to both mechanical and electromagnetic waves.

Examples

1. A stone is dropped into a still pool of water. It generates 20 waves that cover a distance of 10.0 m of the surface. The outer wave covers the 10.0 m in a time of 5.00 s and the average height of the waves is 10.0 mm (crest to trough).

   (a) Determine the wavelength and velocity of the waves.
   (b) Calculate the period of the waves.
   (c) Draw a displacement - time graph to represent the motion of the waves.

(a)

$$\lambda = \frac{10.0}{20} = 0.500 \text{ m} \qquad v\,\cancel{c} = \frac{10.0}{5.00} = 2.00 \text{ ms}^{-1} \qquad v = \frac{s}{t}$$

$\therefore$ Wavelength = 0.500 m and wave velocity = 2.00 ms^{-1}.

(b)

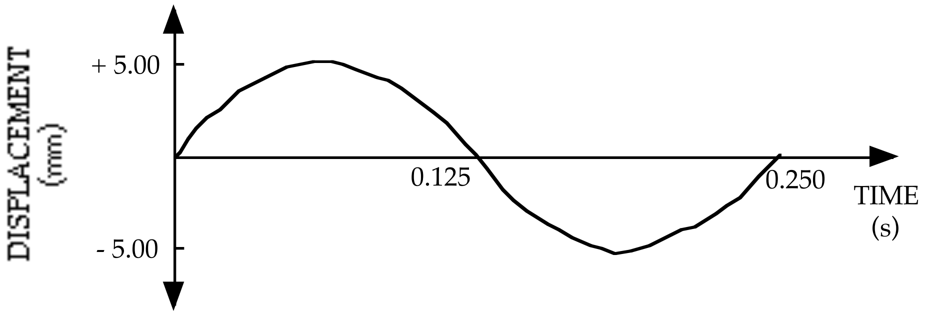
$$v\,\cancel{c} = f\lambda$$

$$\Rightarrow \quad f = \frac{\cancel{c}\,v}{\lambda} = \frac{2.00}{0.500} = 4.00 \text{ Hz}$$

$$T = \frac{1}{f} = \frac{1}{4.00} = 0.250 \text{ s}$$

$\therefore$ Period of the waves = 0.250 s .

(c)



2. Radio station JJJ operates at 99.3 MHz. What is the wavelength of the transmission?

RADIO WAVES ARE PART OF THE LIGHT SPECTRUM

$$c = f\lambda$$

$$\Rightarrow \quad \lambda = \frac{c}{f}$$

$$= \frac{3.00 \times 10^8}{99.3 \times 10^6}$$

$$= 3.021 \text{ m}$$

$\therefore$ Wavelength = 3.02 m .

3.

Waves are created on the surface of the water in a wave pool by a device that dips up and down 20.0 times per minute. The velocity of the resultant wave is 3.00 m s⁻¹.

a State the frequency, period and wavelength of the observed wave.

b What happens to the value of the velocity and wavelength if the frequency of the source is doubled?

Solution

a Since frequency is defined as the number of waves or cycles that pass a given point per second:

$f = 20.0$ per minute

$$f = \frac{20.0}{60} = 0.333 \text{ Hz}$$

$f = 0.333$ Hz

$$T = \frac{1}{f} = \frac{1}{0.333} = 3.00 \text{ s}$$

The period is therefore 3.00 s

The wavelength is then:

$f = 0.333$ Hz

$$v = f\lambda$$

$v = 3.00 \text{ m s}^{-1}$

$$\lambda = \frac{v}{f} = \frac{3.00}{0.333} = 9.00 \text{ m}$$

b The velocity is determined by the properties of the medium and so it is unchanged, $v = 3.00 \text{ m s}^{-1}$. At a set velocity, doubling the frequency will result in wavelengths of half the original value, i.e. 4.50 m.

4.

THE WAVE EQUATION 1

A longitudinal wave has a wavelength of 2.00 m and a speed of 340 m s^{-1}. What is the frequency, f, of the wave?	
Thinking	**Working**
The wave equation states that $v = f\lambda$. Knowing both v and λ, the frequency, f, can be found. Rewrite the wave equation in terms of f.	$v = f\lambda$ $f = \frac{v}{\lambda}$
Substitute the known values and solve.	$f = \frac{v}{\lambda}$ $= \frac{340}{2.00}$ $= 170\text{ Hz}$

5.

THE WAVE EQUATION 2

A longitudinal wave has a wavelength of 2.00 m and a speed of 340 m s^{-1}. Calculate the period, T, of the wave.	
Thinking	**Working**
Rewrite the wave equation in terms of T.	$v = f\lambda$, and $f = \frac{1}{T}$ ← Thinking $v = \frac{1}{T} \times \lambda$ $v = f\lambda$ $= \frac{\lambda}{T}$ $= \frac{\lambda}{T}$ $T = \frac{\lambda}{v}$ $\Rightarrow T = \frac{\lambda}{v}$
Substitute the known values and solve.	$T = \frac{\lambda}{v}$ $= \frac{2.00}{340}$ $= 5.88 \times 10^{-3}\text{ s}$

REFLECTION

Waves are reflected by barriers, the amount of energy reflected being determined by the energy-absorbing nature of the barrier.

Reflecting surfaces can be classified as follows.

Rigid - FIXED END

- Absorbs little or no energy - all is reflected.

    e.g. Walls of a pool are rigid to any water waves.
    String fixed at one end.

- The reflected wave is ***180° out of phase*** to the incident wave.

Fixed and can't move.

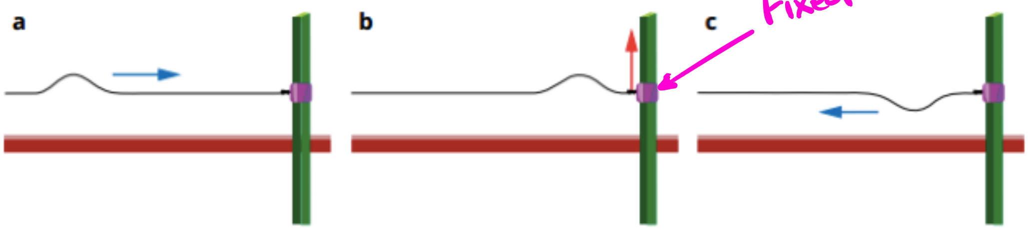


FIGURE 10.3.2 (a) A wave pulse moves along a string to the right and approaches a fixed post. (b) On reaching the end, the string exerts an upwards force on the fixed post. Due to Newton's third law, the fixed post exerts an equal and opposite force on the string which (c) inverts the wave pulse and sends its reflection back to the left on the bottom side of the string. There is a phase reversal on reflection from a fixed end.

Non-Rigid FREE END

- Some energy is absorbed, the rest is reflected.
  i.e. The energy of the reflected wave is ***less*** than the energy of the incident wave.

- The reflected wave is ***in phase*** with the incident wave.

    e.g. Carpet absorbs much of the incident sound energy, resulting in a lower level of reflected sound in a room.

Free to move

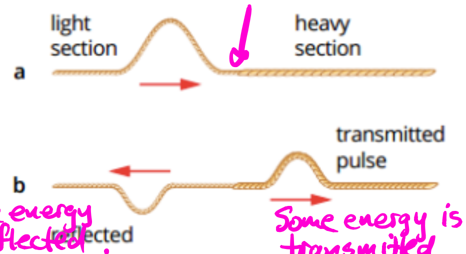


FIGURE 10.3.4 (a) A wave pulse travels along a light rope towards a heavier rope. (b) On reaching a change in density the wave pulse will be partly reflected and partly transmitted. This is analogous to a change in medium.

- A string attached at a "free end" can represent a non-rigid surface.

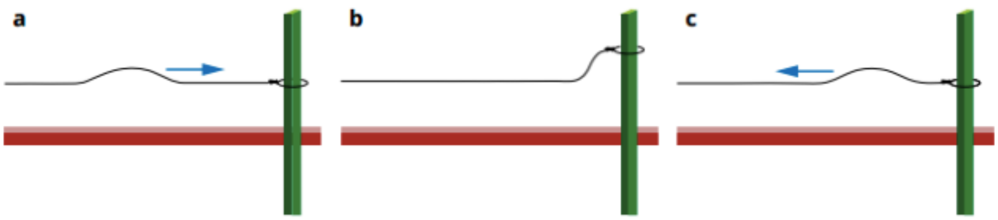


FIGURE 10.3.3 (a) A wave pulse moves along a string to the right and approaches a free end at the post. (b) On reaching the post the free end of the string is free to slide up the post. (c) No inversion happens and the wave pulse is reflected back to the left on the same side of the string, i.e. there is no phase reversal on reflection from a free end.

Laws Of Reflection

1. The angle of incidence (i) = The angle of reflection (r)

2. The incident ray and reflected ray are on opposite sides of the normal, and all three are in the same plane.

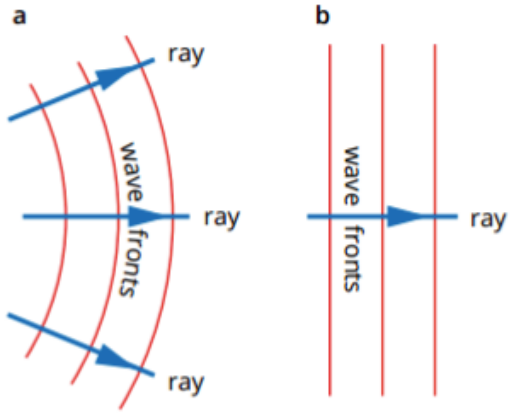


FIGURE 10.3.5 Rays can be used to illustrate the direction of motion of a wave. They are drawn perpendicular to the wave front of a two- or three-dimensional wave and in the direction of travel of the wave; (a) illustrates rays for circular waves near a point source while (b) shows a ray for plane waves.

Remember this

ANGLE OF INCIDENCE = ANGLE OF REFLECTION

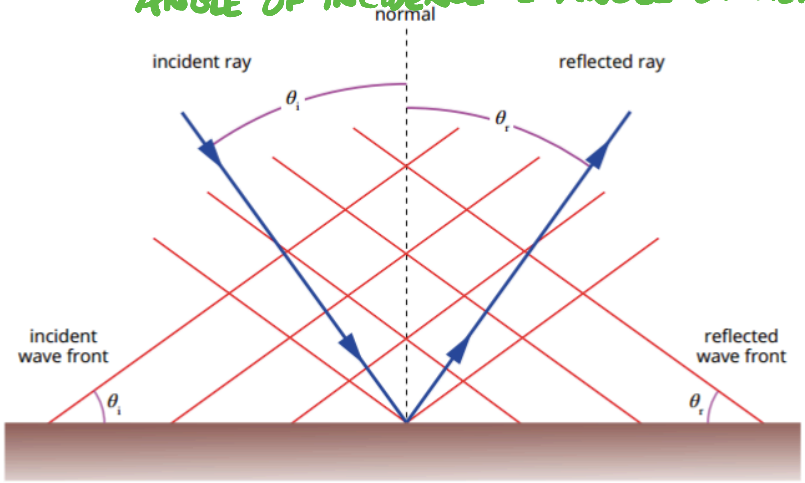


FIGURE 10.3.6 The law of reflection. The angle between the incident ray and the normal (θ_i) is the same as the angle between the normal and the reflected ray (θ_r).

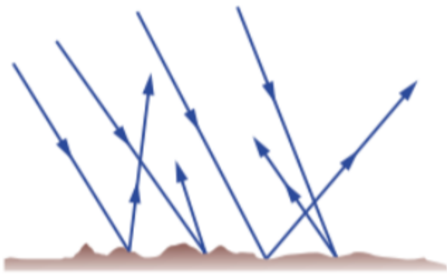


FIGURE 10.3.7 Reflection from an irregular surface. Each incident ray may be reflected in a different direction, depending upon how rough or irregular the reflecting surface is. The resulting wave will be diffuse (spread out).

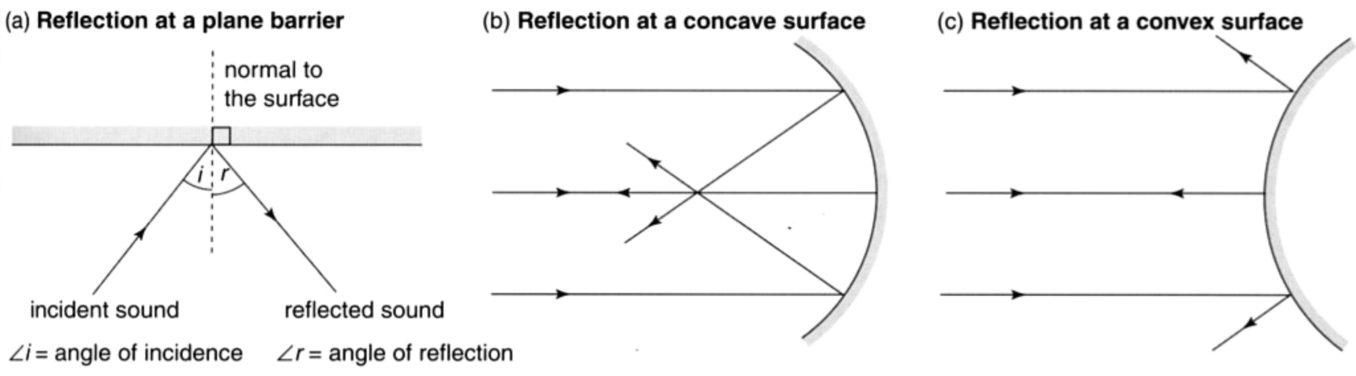
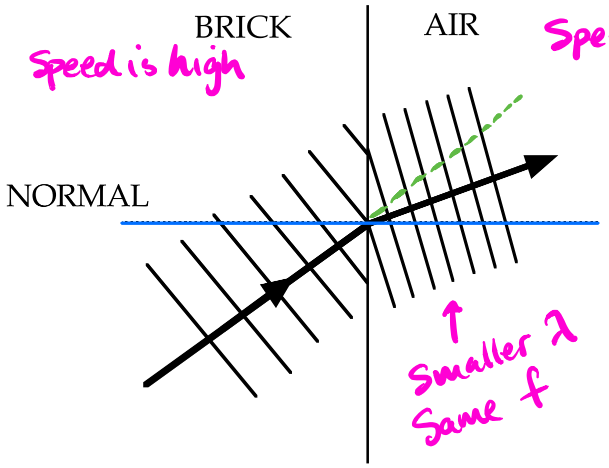


Figure 5.15
Ray tracings of the paths of reflected sound waves.

REFRACTION

Refraction occurs when there is a change in the velocity of a wave as it travels ***obliquely*** from one medium into another.



WAVE SPEED **SLOWS DOWN** IN AIR SO WAVES REFRACT **TOWARDS** THE NORMAL.

SINCE THE FREQUENCY REMAINS CONSTANT, A **SLOWER WAVE** SPEED IMPLIES A **SMALLER WAVELENGTH**.

Slower speed in second medium => smaller λ, ***refract towards normal***.

Faster speed in second medium => larger λ, ***refract away from normal***.

If the incoming wave is ***perpendicular to the boundary of the two media***, ***no refraction occurs***.

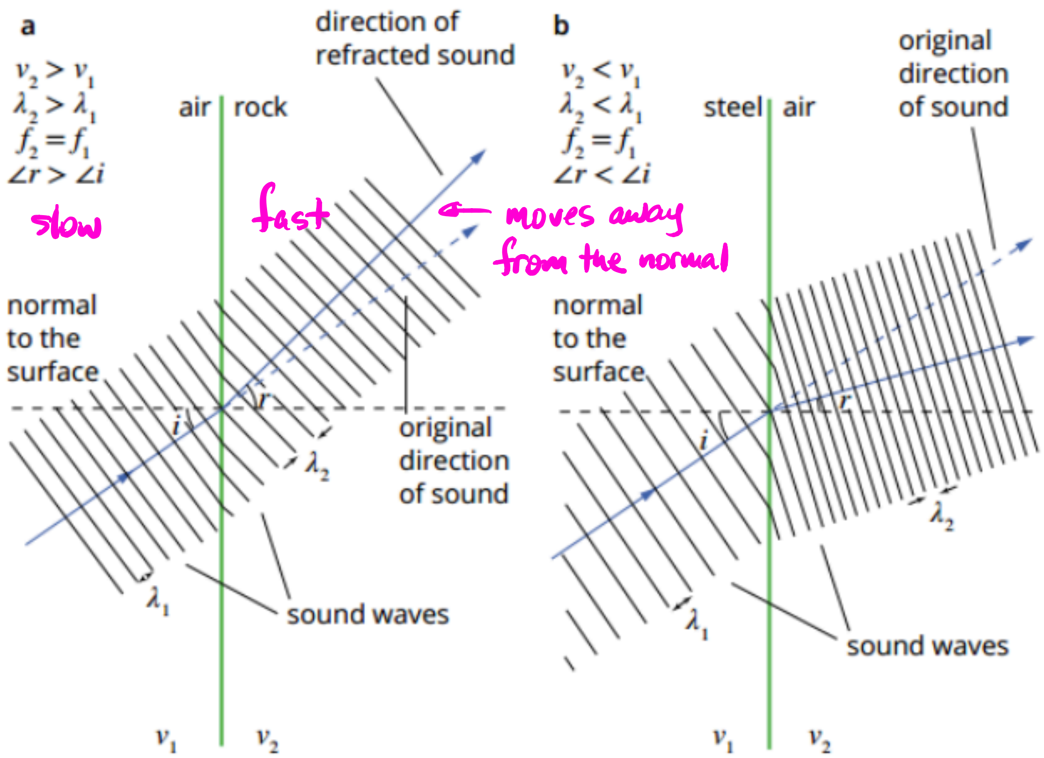
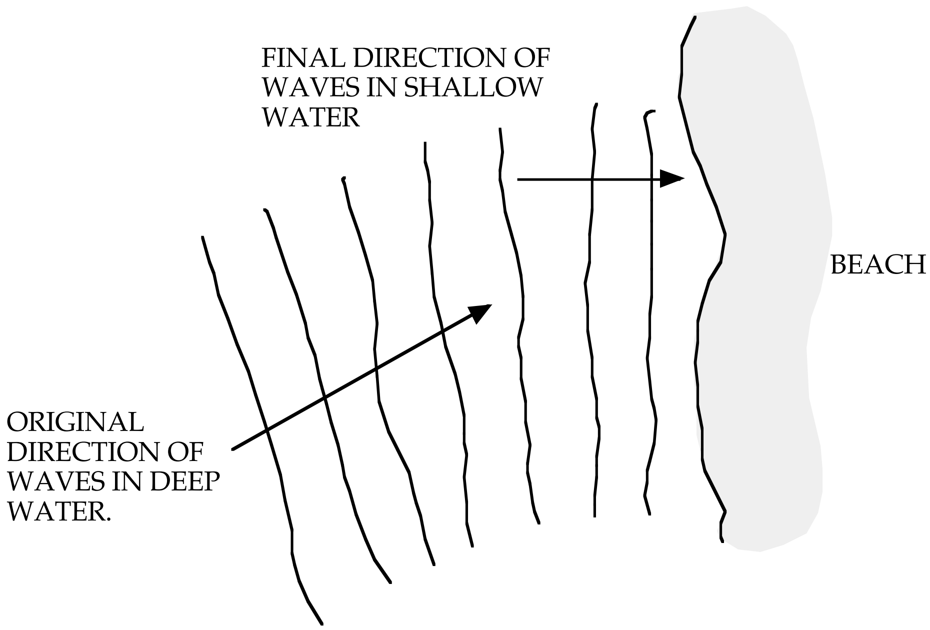


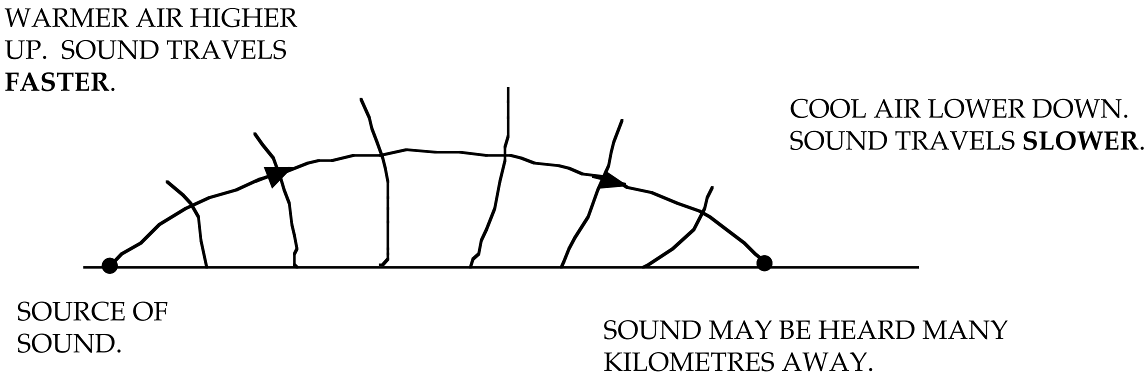
FIGURE 10.3.10 (a) The speed of the wave front in the second medium (the rock) is faster than in the first medium (air) and the wave front bends away from the normal. (b) The speed of the wave front in the second medium (air) is slower than in the first medium (steel) and the wave front bends towards the normal.

Examples

1. Waves in the ocean slow down in shallow water and refract towards a normal to the beach, meaning that the waves arrive nearly parallel to the beach.



2. Sound is refracted by layers of air of differing densities, though the boundaries are ill-defined. When cooler air is trapped near the ground at night, sound can be reflected back down to the ground.



DIFFRACTION

Diffraction may be defined as the spreading out and change in direction of part of a wave as it passes around an obstacle or through an aperture in an obstacle.

It is at a ***maximum*** when the obstacle or aperture is about the ***same size as the wavelength*** of the incident wave.

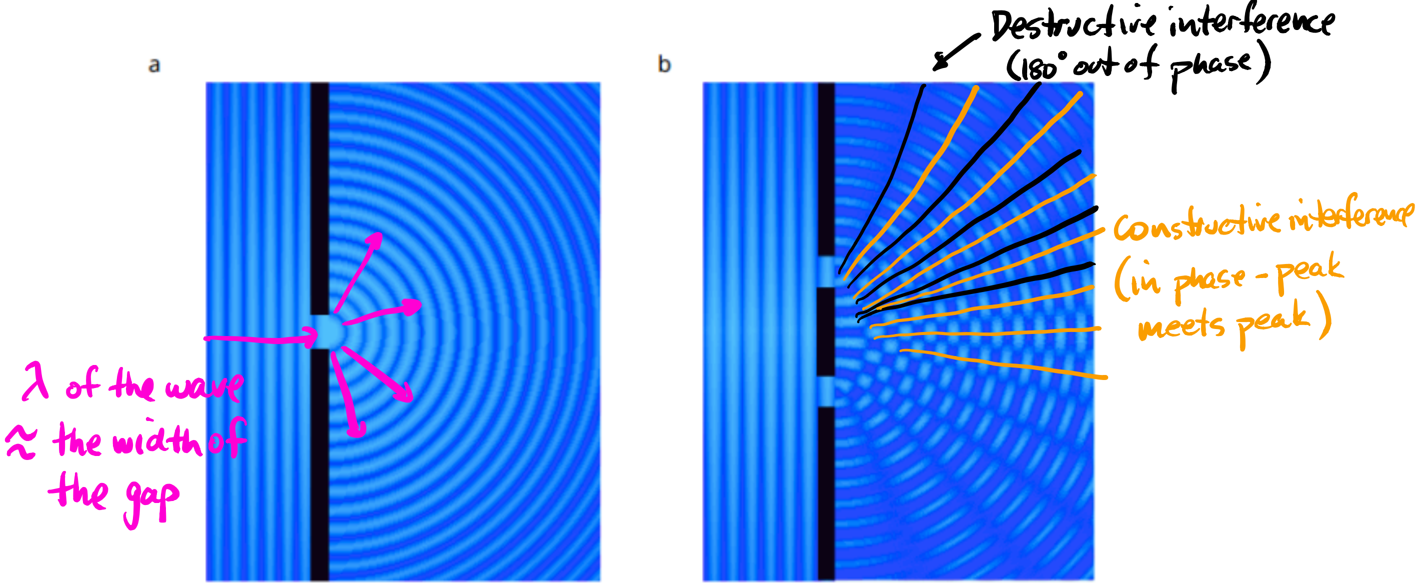


FIGURE 10.3.15 (a) A plane wave undergoes diffraction after passing through an aperture. (b) Interference effects occur due to diffraction from both of the slits.

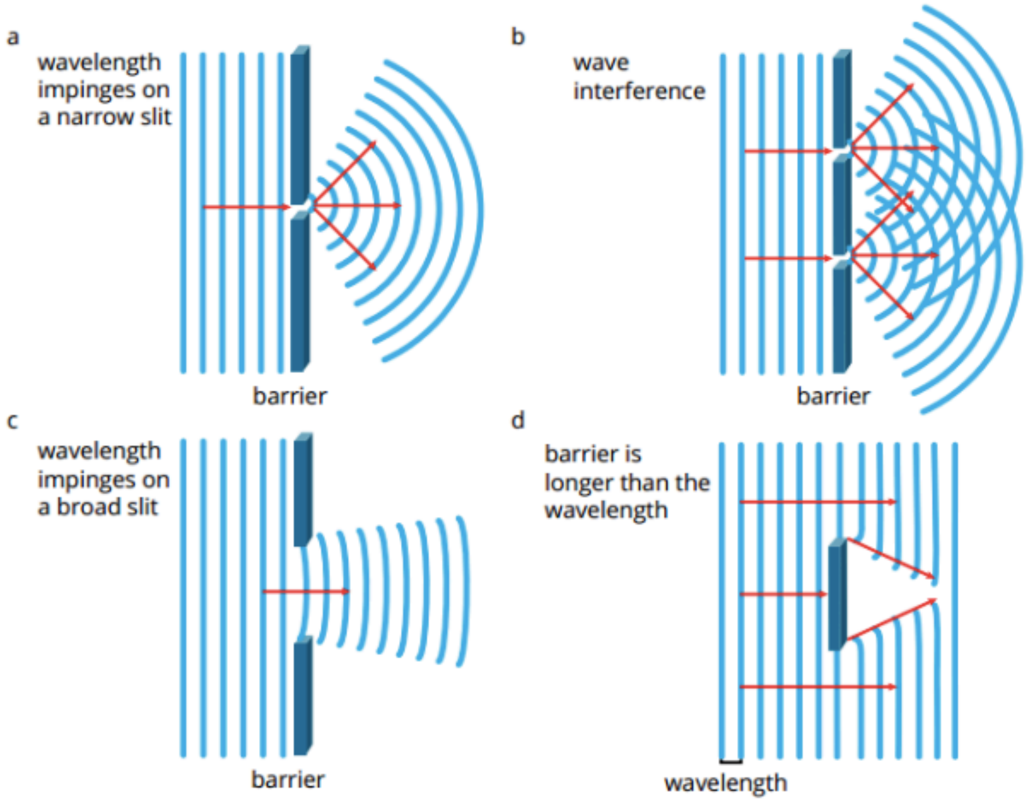
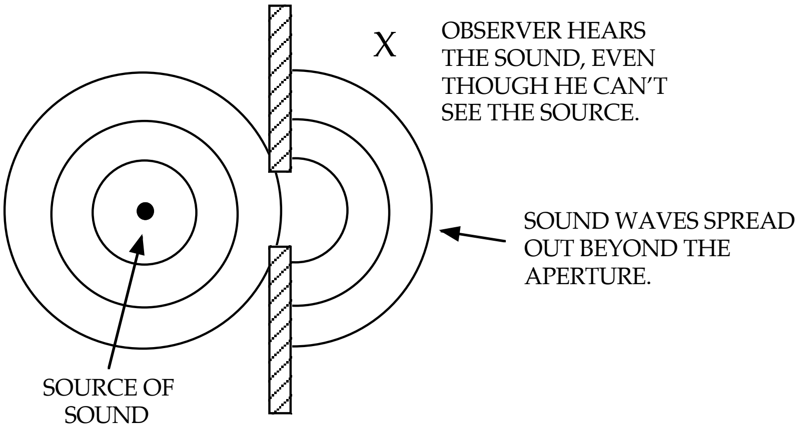


FIGURE 10.3.16 The amount of diffraction depends on the size of the gap or aperture and the wavelength. When the aperture is much smaller than the wavelength, significant diffraction, or spreading out of the waves, will occur ((a) and (b)). A large aperture (c) or a wide barrier (d) relative to the wavelength will cause little diffraction (d). Interference effects occur due to the superposition of two diffracted waves (b).

Examples

1. Sounds made outside a door can quite clearly be heard inside a room. The wavelengths of the sound waves are about the same size as the 1.00 m width of a standard door so the waves diffract into the room.



Reflection of sound from the walls will also help the sound reach the observer.

2. Ships and lighthouses have very low pitch horns so that they are heard clearly. Low sounds have wavelengths of several metres which is typical of distances between objects in a harbour.

3. Visible light waves have wavelengths of 4000 - 7000 Å (4.0×10^{-7} - 7.0×10^{-7} m). This distance is achieved on a ***diffraction grating***, consisting of 600 -1000 lines per millimetre ruled on a piece of glass or plastic. The distance between the lines corresponds to the wavelength of the incident light and diffraction occurs.

4. The wavelength of common sounds means that such sounds can bend around the corner of common objects that are quite large.

   e.g. Sounds can be heard around the corners of buildings.

5.

DIFFRACTION

When high frequency sound waves, for example of 9000 Hz, strike an obstacle such as a person's head, they leave a distinct sound shadow in which little of the sound can be heard. If one ear is closer to the sound source than the other, these higher frequency sounds will be heard as louder by the ear closer to the source.

a Assuming the speed of sound is 340 m s^{-1}, calculate the wavelength of the 9000 Hz sound.

Thinking	Working
The formula is $v = f\lambda$. Therefore $\lambda = \frac{v}{f}$.	$\lambda = \frac{340}{9000}$ $= 3.78 \times 10^{-2}$ m

b Use this calculation to explain why this frequency leaves a sound shadow.

Compare the wavelength of sound to the distance between a person's ears (approximately 20 cm).	$\lambda = 3.78$ cm is much less than 20 cm. Therefore, diffraction effects are minimal.

INTERACTIONS OF WAVES

Principle Of Superposition

When two or more waves travelling through the same medium at the same time meet, their displacements combine.

Waves can interfere ***constructively*** or ***destructively***.

Constructive Interference: two pulses combine to give a pulse of greater amplitude.

(a) Two pulses that are in phase to one another move in opposite directions in the same medium (a string).
(b) When they meet, the resultant is the sum of both amplitudes.
(c) The pulses continue unaffected.

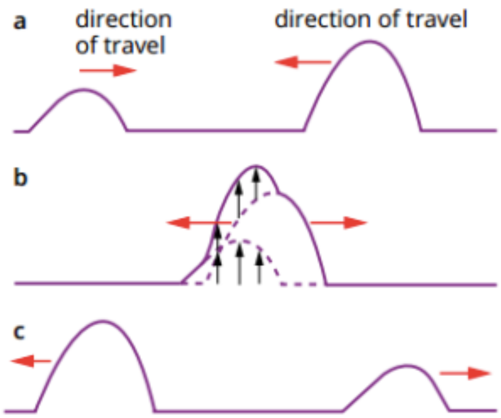


FIGURE 10.4.1 (a) As two wave pulses approach each other superposition occurs. (b) The occurrence of constructive interference. (c) After the interaction, the pulses continue unaltered; they do not permanently affect each other.

Destructive Interference: two pulses combine to produce a pulse with a smaller amplitude than either of the two original amplitudes.

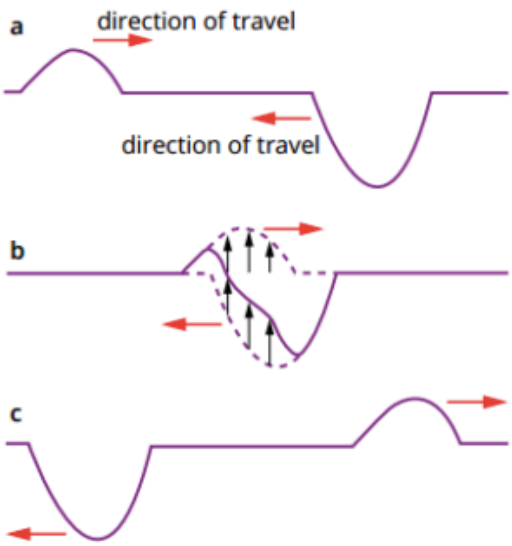


FIGURE 10.4.2 (a) As two wave pulses approach each other superposition occurs. (b) Superposition of waves in a string showing destructive interference. (c) As in constructive interference, the waves do not permanently affect each other.

(a) Two pulses 180° out of phase to each other are moving in the same medium.
(b) Both pulses cancel to give a zero resultant.
(c) The pulses continue unaffected.

Stationary Waves

These are produced when two identical waves of ***equal amplitude, wavelength and speed*** travel in ***opposite directions*** in the same medium.

The resulting wave has points of zero displacement (called ***nodes***) and maximum displacement (called ***antinodes***) occurring in the same position.

i.e. The wave is not "travelling" but "stationary".

Example

Plucking a string on a guitar (or any string instrument) will send a pulse down it to a fixed end. Here it reflects 180° out of phase and moves back down the string to the opposite end (which is also fixed).

A stationary (or standing wave) is produced with its nodes and antinodes appearing not to move.

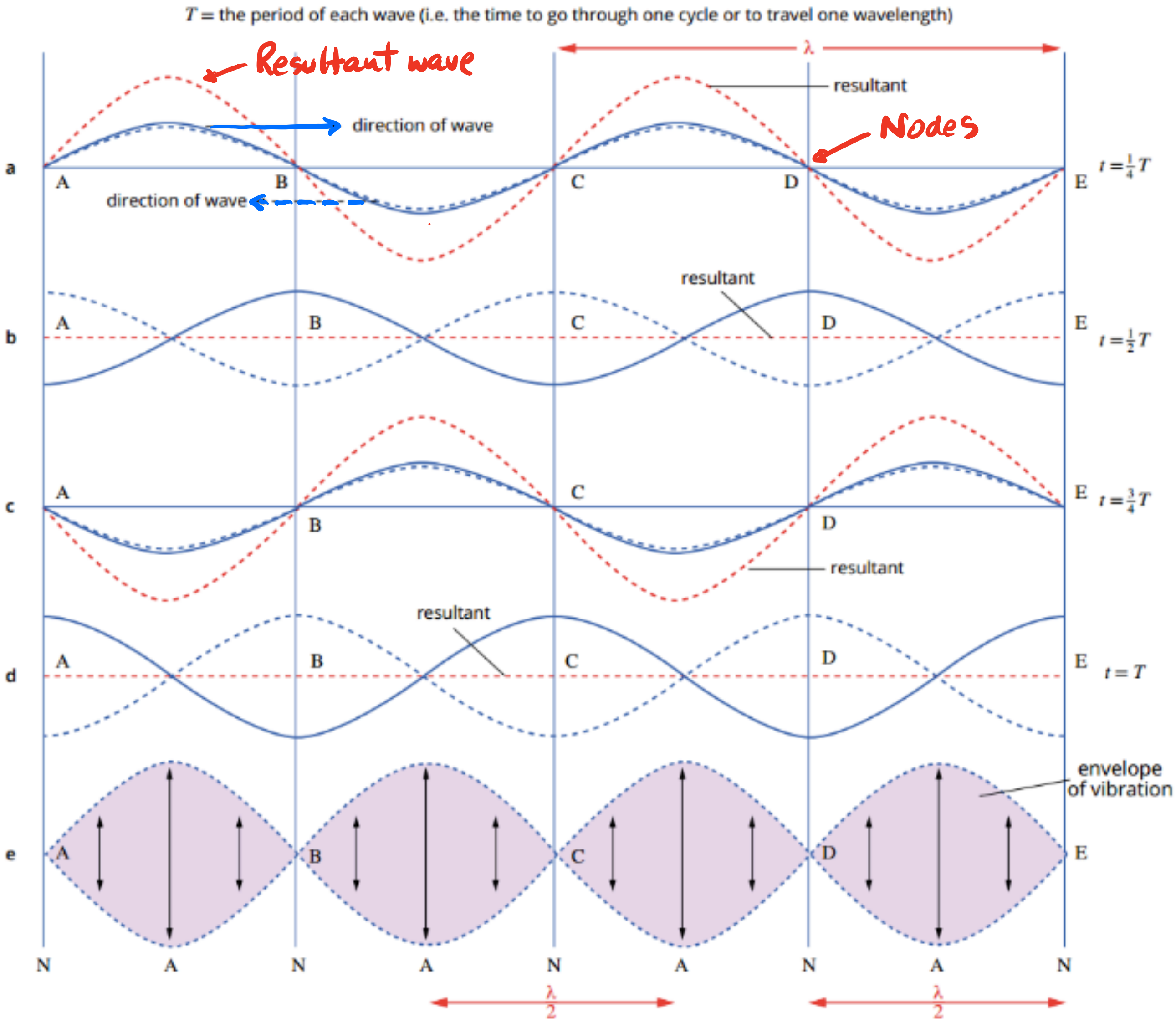


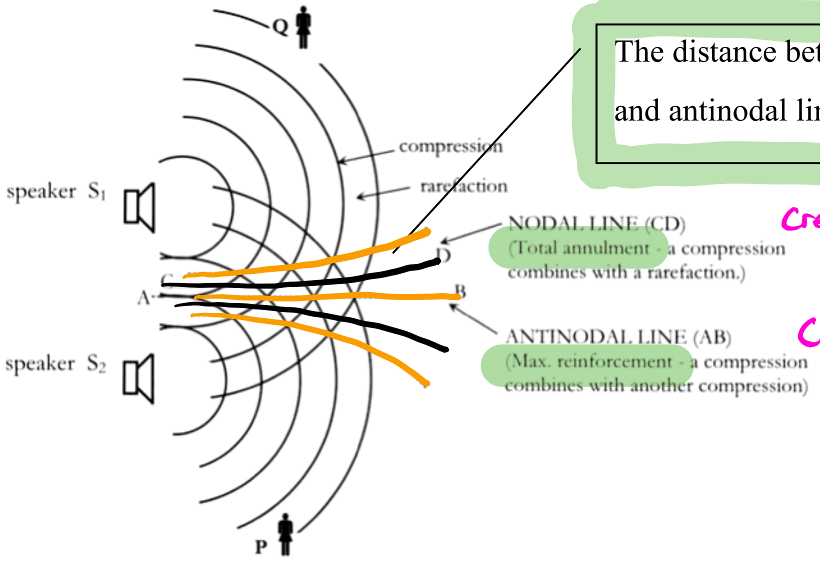
FIGURE 10.4.3 (a) Complete constructive interference occurs when two waves of the same amplitude that are exactly in phase add together to give a wave of twice the amplitude. (b) Complete destructive interference occurs when the two waves that are 180° or $\frac{\lambda}{2}$ out of phase add together to completely cancel out and give a wave of zero amplitude.

Interference From Two Point Sources

Consider two loudspeakers emitting the same sound (in phase) from two different points.

Points of maximum reinforcement (***antinodes***) and total annulment (***nodes***) will occur as shown in the diagram below.

Important!

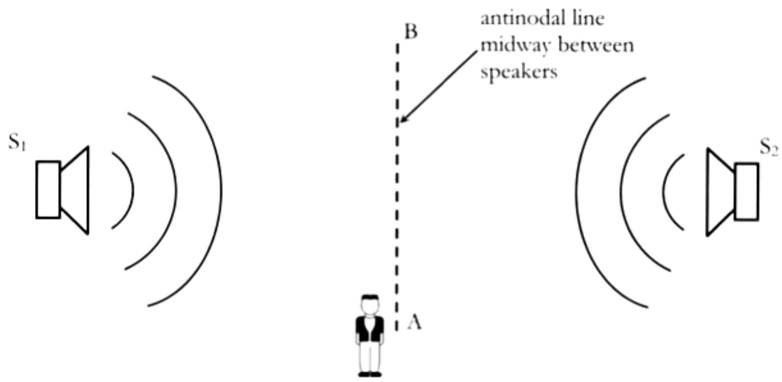


The distance between a nodal and antinodal line $= \frac{\lambda}{2}$.

Crest meets trough (180° out of phase)

Crest meets crest

A similar interference effect (nodes and antinodes) will occur when two loudspeakers are facing each other and emitting the same sound in phase. A standing wave pattern forms between the speakers.



- ***Antinodal lines*** occur where there is ***constructive*** interference - a ***loud sound*** is heard.
- ***Nodal lines*** occur where there is ***destructive*** interference - a ***soft sound*** is heard.

The difference in distance from either speaker is called the ***path difference***. There are two relationships to describe when constructive and destructive interference occur.

Constructive Interference: $\boxed{\text{path difference} = n\lambda}$ Must arrive in phase

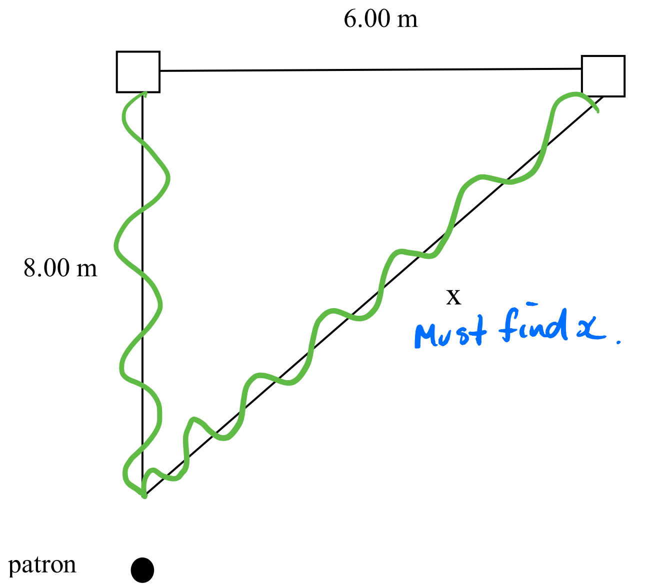
where $n = 0, 1, 2, 3, \ldots$

Destructive Interference: $\boxed{\text{path difference} = \frac{n\lambda}{2}}$ Must arrive $\frac{\lambda}{2}$ out of phase

where $n = 1, 2, 3, \ldots$

Examples

1. In a cinema, a patron is sitting in front of two large speakers as shown. Assume the speakers are in phase with each other.



(a) Calculate the path difference of the sound heard by the patron.

$$x = \sqrt{(6.0)^2 + (8.0)^2}$$
$$= 10.0 \text{ m}$$

$$\therefore \quad \text{path difference} = 10.0 - 8.0$$
$$= 2.0 \text{ m}$$

Calculate λ (b) If the speakers are both producing 680 Hz, what does the patron hear?

$$v = f\lambda$$
$$\Rightarrow \quad \lambda = \frac{v}{f}$$
$$= \frac{340}{680}$$
$$\boxed{\lambda = 0.50 \text{ m}}$$

path difference $= 2.0$ m $= 4 \times 0.5$

$= 4\lambda$ ← If this $4\frac{1}{2}\lambda$, it is destructive

$\Rightarrow$ constructive interference

$\therefore$ Patron hears a loud sound.

(c) What is the lowest frequency needed for a soft sound?

$$\text{Lowest frequency} \Rightarrow \text{Large } \lambda$$

$$\text{Minimum path difference} = \frac{\lambda}{2} = 2.0 \text{ m}$$

$$\Rightarrow \lambda = 4.0 \text{ m}$$

$$v = f\lambda$$

$$\Rightarrow f = \frac{v}{\lambda} = \frac{340}{4.0} = \underline{85.0 \text{ Hz}}$$

2.

Two loudspeakers (A and B) are set up as shown in Figure 1.14(a) so that they both emit sounds of 425 Hz which are in phase. The speakers are 2.40 m apart. Assume the speed of sound is 340 ms^{-1}.

(a) What is the wavelength of the sounds emitted?

(b) Melissa stands exactly midway between the speakers and then walks away so as to always remain equidistant from them. Describe what she hears. Explain why.

(c) Melissa now walks to a point C, 7.0 m directly in front of speaker A.

(i) How far is she from speaker B?

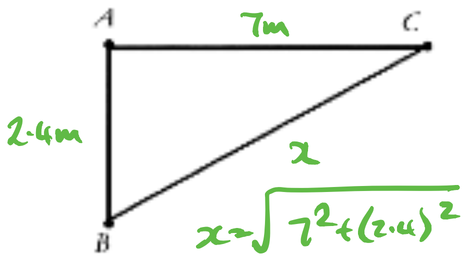
(ii) What will be the effect?

$v = 340 \text{ ms}^{-1}$

$f = 425 \text{ Hz}$

$\lambda = ?$

$AB = 2.40 \text{ m}$



$x = \sqrt{7^2 + (2.4)^2}$

$= 7.4 \text{m}$

(a) $v = \lambda f \quad \therefore \lambda = \frac{v}{f} = \frac{340}{425}$

$= 0.80 \text{ m}$

(b) If Melissa is always equidistant from the speakers the sounds reaching her will always be in phase. She will therefore always hear a loud sound.

(c) $AB = 2.40 \text{ m}$ $\qquad (BC)^2 = (AC)^2 + (AB)^2$

$AC = 7.00 \text{ m}$ $\qquad = 49 + 5.76$

$BC = ?$ $\qquad \therefore BC = 7.40 \text{ m}$

Since the wavelength of the sound is 0.80 m then at point C the sounds are exactly half a wavelength apart (180° out of phase). Melissa will "hear" silence.

$\lambda = 0.8\text{m} \quad \therefore \# \lambda = \frac{7.4}{0.8} = 9.25\lambda \quad \therefore$ Destructive

PRODUCTION OF SOUND

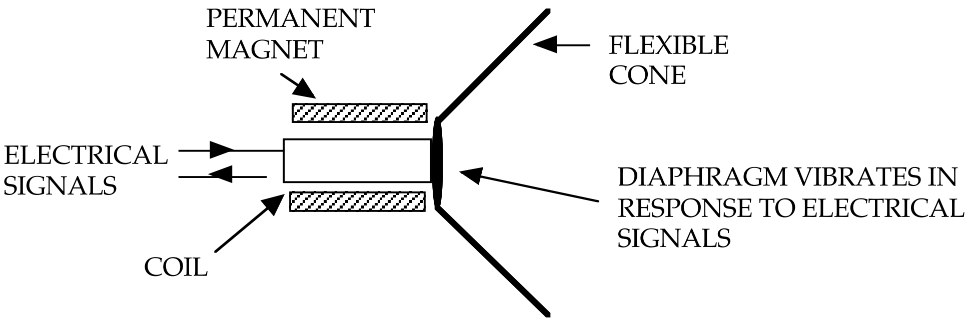
A sound wave can be generated by a vibrating object imparting a longitudinal wave motion to the surrounding medium.

The particles of the medium receive energy from the vibrating source and begin to vibrate at the same frequency.

The two fundamental requirements for the production and propagation of sound are a vibrating source and an elastic medium.

Example

A loudspeaker.



The coil moves in response to electrical signals as per the motor principle - its magnetic field interacts with the permanent magnet's field.

The diaphragm is connected to the coil and moves accordingly.

Sound is produced by the diaphragm vibrating, causing the cone to push against the air particles and transfer energy to them.

As the cone moves forwards, a compression is produced. As it moves backwards, a rarefaction is produced.

SPEED OF SOUND

The velocity of sound in a medium is determined by the ***elasticity*** and ***density*** of a medium.

The greater the elasticity, the greater the velocity.

The greater the density, the slower the velocity.

The velocity of sound in air is affected by temperature but not pressure.

i.e. Increasing temperature increases the velocity.

FREQUENCY AND PITCH

Increasing the frequency increases the pitch of the sound.

The pitch of a sound is the ***perception*** a listener has of a sound - whether it is high or low.

The human ear has a frequency range of 20 - 20,000 Hz.

Frequencies above 20,000 Hz are called ***ultrasonic***. Dogs hear up to 50,000 Hz while bats respond up to 100,000 Hz.

Those less than 20 Hz are called ***infrasonic***. These frequencies are generated by earthquakes and heavy machinery vibrating.

CHARACTERISTICS OF SOUND WAVES

Sounds can be classified as either musical notes or noise.

> ***Musical Notes***: vibrating body produces a regular frequency and a definite pitch.
>
> ***Noise***: produced by a body that does not vibrate with a regular frequency.

Stretched strings and columns of air vibrate in particular ways to produce sound.

Vibrations In Stretched Strings

As seen before, a standing wave is set up in a string that is attached at both its ends.

A string will vibrate in different ways according to how it is plucked, strummed, struck or where a bow is pulled across it.

> i.e. In the middle, down towards one end, etc.

The lowest frequency at which a string will vibrate is called the ***fundamental frequency*** (or ***1st harmonic***).

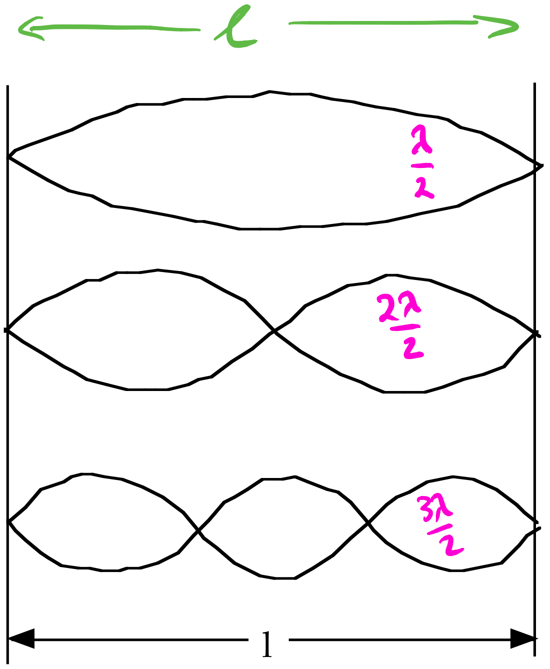
The next mode of vibration is called the ***second harmonic*** and has a ***frequency that is twice that of the fundamental frequency***.

Subsequent harmonics have frequencies that are whole number multiples of the fundamental frequency.

> ***Harmonic***: A whole number multiple of the fundamental frequency.

Fundamental

Draw these ALWAYS



HARMONIC	ℓ	f
1st	$1/2\,\lambda$ $\frac{\lambda}{2}$	f_1 $1 \times f_1$
	$\ell = \frac{\lambda}{2}$ $\Rightarrow \lambda = 2 \times \ell$	
2nd	$2/2\,\lambda$ $\frac{2\lambda}{2}$	$2f_1$ $2 \times f_1$
3rd	$3/2\,\lambda$ $\frac{3\lambda}{2}$	$3f_1$ $3 \times f_1$
4th	$\frac{4\lambda}{2}$	

NOTICE THAT THE HARMONICS HAVE WHOLE-NUMBERED MULTIPLES OF THE FUNDAMENTAL LENGTH (l) AND FREQUENCY (f).

In strings, **all** of the harmonics are possible.

i.e. n = 1, 2, 3, 4, ...

The frequency of the harmonics in a string is given by:

$$v = \frac{f 2\ell}{n} \leftarrow \boxed{f = \frac{nv}{2l}}$$

On data sheet

where
f = frequency of vibration (Hz).
n = number of the harmonic.
v = speed of sound (ms^{-1}).
l = length of the string (m).

Consider the 3rd harmonic

$\ell = \frac{3\lambda}{2}$ $f = 200\,Hz$, $\ell = 1{\cdot}0\,m$

$\Rightarrow \lambda = \frac{2\ell}{3} = \_\_\_ = \_\_\_ m$

$\therefore v = f\lambda$

$v = \frac{f 2\ell}{3} = \frac{(200)\,2\,(1{\cdot}0)}{3}$

$= \underline{133\ ms^{-1}}$

Examples

1. When the key on a piano corresponding to a note of 440 Hz is struck firmly, it is found that the 880 Hz string also vibrates, even though the two are not connected and are some distance apart. How can this be so?

Solution

This situation involves resonance. The frequency of the 880 Hz string corresponds to the second harmonic of the forcing vibration by the 440 Hz string (i.e. it is double the frequency). Hence, it will resonate at its natural frequency of vibration. The sound produced by the 440 Hz string provides the necessary energy.

2.

FUNDAMENTAL FREQUENCY

A violin string, fixed at both ends, has a length of 22 cm. It is vibrating at its fundamental frequency of vibration, at a frequency of 880 Hz.

a Calculate the wavelength of the fundamental frequency.	
Thinking	**Working**
Identify the length of the string (ℓ) in metres and the harmonic number (n).	$\ell = 22\text{ cm} = 0.22\text{ m}$ $n = 1$
Recall that for any frequency, $\lambda = \frac{2\ell}{n}$. Substitute the values from the question and solve for λ.	$\lambda = \frac{2\ell}{n}$ $= \frac{2 \times 0.22}{1}$ $= 0.44\text{ m}$ $\ell = \frac{\lambda}{2} \Rightarrow \lambda = 2\ell$ $= 2 \times 0.22$ $= 0.44\text{m}$

b Calculate the wavelength of the second harmonic.	
Thinking	**Working**
Identify the length of the string (ℓ) in metres and the harmonic number (n).	$\ell = 22\text{ cm} = 0.22\text{ m}$ $n = 2$
Recall that for any frequency, $\lambda = \frac{2\ell}{n}$. Substitute the values from the question and solve for λ.	$\lambda = \frac{2\ell}{n}$ $= \frac{2 \times 0.22}{2}$ $= 0.22\text{ m}$ $\ell = \frac{2\lambda}{2} \Rightarrow \lambda = \ell$ $= 0.22\text{m}$

Air Columns

The air in an open or closed pipe will resonate to a vibrating source placed at one end.

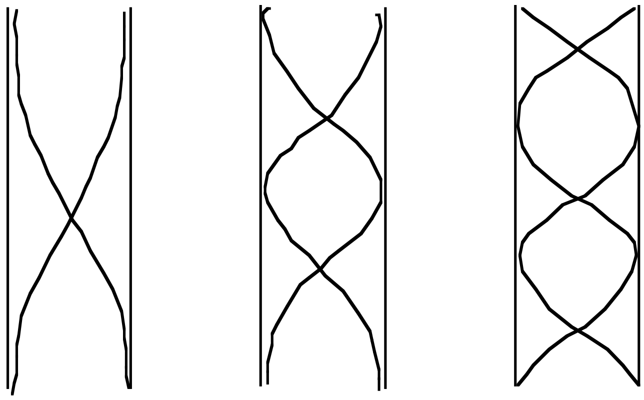
> e.g. A tuning fork, blowing across the top of the pipe (the turbulence causes the vibrations), a vibrating reed in a clarinet.

A longitudinal standing wave is set up in the pipe with reflection occurring from each end, regardless of whether it is the open or closed end.

At an ***open end,*** only an ***antinode*** is possible, and at a ***closed end*** only a ***node*** is possible.

Open Pipes

Draw these



HARMONIC	1st	2nd	3rd
l	$1/2\,\lambda$ $\frac{\lambda}{2}$	λ $\frac{2\lambda}{2}$	$3/2\,\lambda$ $\frac{3\lambda}{2}$
f	f_1 $1 \times f_1$	$2f_1$ $2 \times f_1$	$3f_1$ $3 \times f_1$

For open pipes, ***all harmonics are possible***.

7th harmonic $l = 2.0\text{ m}$ $f = 200\text{ Hz}$

$$l = \frac{7\lambda}{2}$$

$$\Rightarrow \lambda = \frac{2l}{7} = \frac{2(2.0)}{7}$$

$$= \_\_\_\_\_\_ \text{ m}$$

The fundamental length of the pipe is $l = 1/2\,\lambda$.

The distance between successive harmonic lengths is $1/2\,\lambda$.
The frequency of the harmonics in an open pipe is given by:

$$\boxed{f = \frac{nv}{2l}}$$

where
- f = frequency of vibration (Hz).
- n = number of the harmonic.
- v = speed of sound (ms^{-1}).
- l = length of the open pipe (m).

$$v = f\lambda$$
$$= (200)(\quad)$$
$$= \_\_\_\_\_\_ \text{ ms}^{-1}$$

Examples

1. An open pipe has an effective length of 1.23 m. If the speed of sound in air is 3.40×10^2 ms^{-1}, calculate:

   (a) the fundamental frequency.
   (b) the frequency and wavelength of the third harmonic.

(a) For the first harmonic:

$$l = 1/2\,\lambda$$

$$\Rightarrow \quad \lambda = 2(1.23) = 2.46 \text{ m}$$

$\lambda = 2\ell$

$$c = f\lambda$$

$$\Rightarrow \quad f = \frac{c}{\lambda} = \frac{3.40 \times 10^2}{2.46} = 1.382 \times 10^2 \text{ Hz}$$

$\therefore$ Fundamental frequency = 1.38×10^{2} Hz.

(b) For the third harmonic:

$$l = 3/2\,\lambda$$

$$\Rightarrow \quad \lambda = 2/3(1.23) = 0.820 \text{ m}$$

$\ell = \frac{3\lambda}{2}$

$\lambda = \frac{2\ell}{3}$

$= .\,- - -$

$$f_3 = 3f_1 = 3(1.38 \times 10^2) = 4.14 \times 10^2 \text{ Hz}$$

$\therefore$ Wavelength = 0.820 m and frequency = 4.14×10^{2} Hz.

2.

OPEN-ENDED AIR COLUMNS

A particular flute has an effective length of 35 cm. It can be thought of as an open-ended air column. The speed of sound is 340 m s^{-1}.

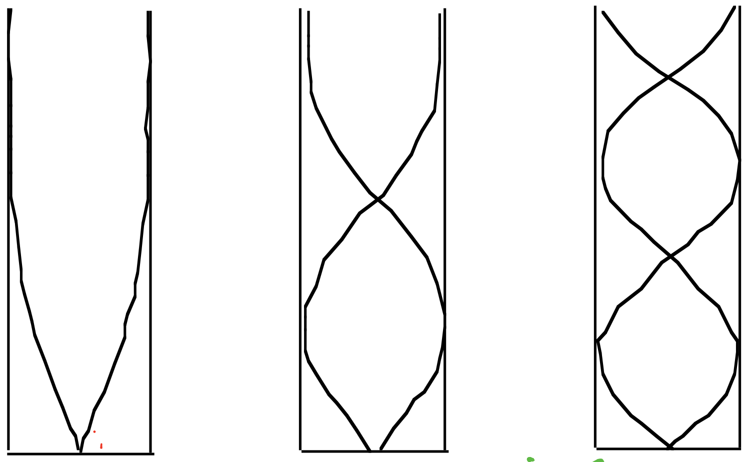
a Calculate the wavelength of the second harmonic.	
Thinking	**Working**
Identify length of the air column (ℓ) in metres and the harmonic number (n).	$\ell = 35$ cm $= 0.35$ m $n = 2$
Recall that for any frequency, $\lambda_n = \frac{2\ell}{n}$. Substitute the values from the question and solve for λ.	$\lambda_n = \frac{2\ell}{n}$ $\lambda_2 = \frac{2 \times 0.35}{2}$ $= 0.35$ m

b Determine the frequency of the second harmonic.	
The frequency can be calculated using $f_2 = \frac{v}{\lambda_2}$	$f_2 = \frac{340}{0.35}$ $= 971$ Hz

c Determine the frequency of the fourth harmonic.	
Recall $f_n = nf_1$	$f_4 = 4f_1$ $= 2f_2$ $= 2 \times 971$ $= 1940$ Hz

Closed Pipes

Draw these



HARMONIC	1st	diff = $\frac{\lambda}{2}$	3rd	diff = $\frac{\lambda}{2}$	5th
l	1/4 λ $\frac{\lambda}{4}$		3/4 λ $\frac{3\lambda}{4}$		5/4 λ $\frac{5\lambda}{4}$
ν	ν1		3ν1		5ν1

Remember the fundamental

For closed pipes, ***only odd harmonics occur***.

The fundamental length is l = 1/4 λ.

The distance between successive harmonic lengths is 1/2 λ.

The frequency of the harmonics in a closed pipe is given by:

$$\boxed{f = \frac{nv}{4l}}$$

7th harmonic $\quad l = 1{\cdot}0\,m \quad f = 300\,Hz$

$$l = \frac{7\lambda}{4}$$

$$\Rightarrow \lambda = \frac{4l}{7} = \frac{4(1{\cdot}0)}{7} = \_\_\_\_ m$$

$$v = f\lambda$$
$$= (300)(\quad)$$
$$= \_\_\_\_\_ ms^{-1}$$

Examples

1.

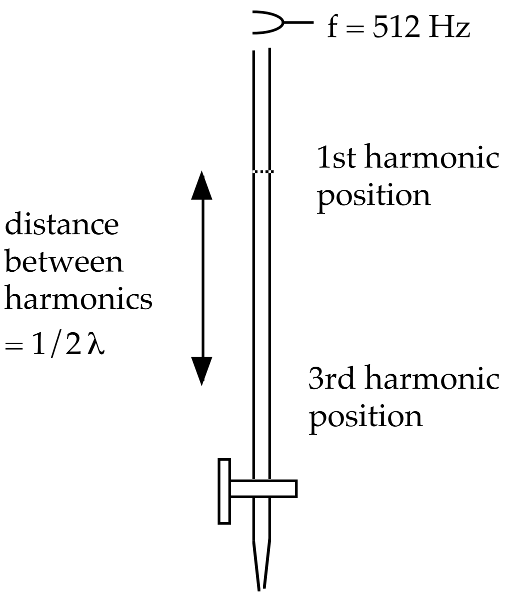
AIR COLUMN CLOSED AT ONE END

The ear canal, from the outer ear to the eardrum, can be thought of as a tube closed at one end (by the eardrum) and open at the other. It is approximately 3.0 cm long in an adult.	
a Calculate the wavelength of the fundamental frequency of the adult ear canal.	
Thinking	**Working**
Identify the length of the air column (ℓ) in metres and the harmonic number (n).	$\ell = 3$ cm $\ell = 0.03$ m $n = 1$
Recall that for any frequency for closed tubes, $\lambda_n = \frac{4\ell}{n}$. Substitute the values from the question and solve for λ.	$\lambda_n = \frac{4\ell}{n}$ $= \frac{4 \times 0.03}{1}$ $= 0.12$ m

b Assuming that the speed of sound through air is 340 m s^{-1}, what frequency does this wavelength correspond to?	
Thinking	**Working**
Identify the speed of the sound (v) in m s^{-1} and the wavelength (λ) from the previous question.	$v = 340$ m s^{-1} $\lambda_1 = 0.12$ m
Recall the wave equation $v = f\lambda$. Rearrange to find f.	$f = \frac{v}{\lambda_1}$ $= \frac{340}{0.12}$ $= 2833$ Hz ≈ 3 kHz

2. A group of students performed an experiment in the laboratory to determine the velocity of sound in air. They filled a burette with water and sounded a 512 Hz tuning fork over the top of it as they let the water slowly flow out the bottom. They found two successive resonant positions when the air column was 0.170 m and 0.505 m from the top.

Using this information, calculate the value for the speed of sound the students obtained.



The first position is not used for calculating the speed of sound since there is an "end effect" that has to be taken into account. This is due to the particles near the open end being able to move in all directions rather than one dimension.

The distance between the two resonant positions is equal to $1/2 \lambda$.

$$\text{i.e.} \quad 0.335 = 1/2\lambda$$
$$\Rightarrow \quad \lambda = 2(0.335)$$
$$= 0.670 \text{ m}$$

$$c = f\lambda$$
$$= (512)(0.670)$$
$$= 3.430 \times 10^2 \text{ms}^{-1}$$

∴ The speed of sound = 3.43×10^2 ms^{-1}.

Musical Sounds

Musical sounds have three distinct characteristics - ***pitch, loudness and quality***.

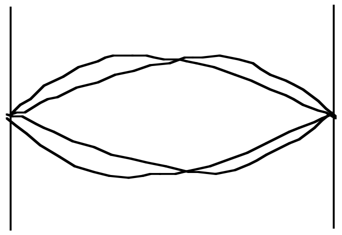
When a piano and a flute play the same note having the same pitch and loudness, there is a clear difference between the sound produced by each instrument. There is a ***difference in the quality*** of the sounds.

The quality of a musical note depends on the ***number, nature and relative intensities of the harmonics*** present in the note.

The greater the number of harmonics present in a note, the better its quality.

Harmonics depend on the mode of vibration of a stretched string or an air column.

e.g. Consider a stretched string.



A STRETCHED STRING VIBRATING WITH ITS FUNDAMENTAL AND SECOND HARMONIC SIMULTANEOUSLY.

Physics in action — Music or noise?

What is music to one person may be just plain noise to another. The frequencies that are present in sounds vary, and any clear distinction between music and noise is mainly a subjective judgement. Physics can provide a basic distinction between what we define as a harmonious sound and as noise, but it does not attempt to draw a line between the two.

Related and unrelated frequencies

Musical instruments generally produce sounds which are whole-number multiples of a fundamental frequency. When a sound from a musical instrument is displayed on an oscilloscope or a digitiser, a clear and distinctive stable wave pattern is seen. This is because the vibrations produced by most musical instruments are only free to vibrate in one dimension. A sound which includes these simple multiples of the fundamental frequency is usually regarded as harmonious.

Hitting two sticks together creates a different mixture of frequencies. The vibrations within the sticks that cause the sound move in three dimensions, and the range of frequencies produced is more complex. The display on a digitiser or an oscilloscope would not show a stable wave pattern; the frequencies bear little or no relation to each other. The sound may still be quite pleasant to hear, but it can be interpreted as noise since it consists of a random mixture of frequencies. 'White noise' is such a mixture of frequencies. Although we define it as noise, it can be soothing and is quite a pleasant sound to many people.

Instruments such as drums lie between the extremes of simple multiples of frequencies and random mixtures. Drums can vibrate in two dimensions, so they do not have the pure harmonious sound of other instruments, but the frequencies do bear some relation to each other. Totally enclosed drums can still sound harmonious, because certain frequencies resonate and are accentuated.

Harmonies and musical scales

Harmonies are an important part of music. Two notes generally sound harmonious if the ratio of their frequencies is a simple whole number. The simplest ratio of 2:1 sounds the most harmonious: this is what we call an octave. A ratio of 3:2, called a 'fifth' since it includes five musical intervals (tones and semitones), also sounds very harmonious. Combinations based on more complex ratios, such as 5:4, sound 'dissonant'. Modern music has seen the acceptance of more dissonant combinations, and today almost any dissonance can be found in a musical score. Many different musical scales involving these ratios have been used throughout the world. For example, African music uses much smaller intervals between notes and thus has two to four times as many defined notes as traditional Western music.

Figure 5.37
Because the vibration of a drum skin occurs in two dimensions, in contrast to the single longitudinal dimension of a string, it creates less harmonious sounds than many other musical instruments.

The most common modern scale is called the 'equally tempered chromatic scale'. This evolved from a scale based on four intervals corresponding to simple fractions of the length of a vibrating string. Various versions of the earliest chromatic scale led to a compromise that suits keyboard instruments such as the piano. (Earlier versions had many more intervals within one octave and would have required a large and unplayable keyboard.) The equally tempered chromatic scale divides a simple 2:1 ratio, one octave, into 12 equal semitones. Each pair of adjacent notes thus has the same frequency ratio.

As an octave is a basic ratio of 2:1, having 12 equal intervals means that the ratio of the frequency of any one note on a piano to that of the next will be $\sqrt[12]{2}:1$ or about 1.06:1.

table 5.1 Frequencies of common notes on the equally tempered chromatic scale, based on A = 440 Hz The tonal names refer to the seven white notes on the keyboard. Compromises in the scale mean that adjacent sharps (#) and flats (b) have the same frequency. (Upper C actually has exactly double the frequency of middle C. It is noted here as 523 and not 524 Hz because the values are rounded to the nearest whole number.)

Note	Tonal name	Frequency (Hz)
A		220
$A^{\#}$, B^{b}		233
B		247
C (middle)	do	262
$C^{\#}$, D^{b}		277
D	re	294
$D^{\#}$, E^{b}		311
E	mi	330
F	fa	349
$F^{\#}$, G^{b}		370
G	so	392
$G^{\#}$, A^{b}		415
A	la	440
$A^{\#}$, B^{b}		466
B	ti	494
C	do	523

FORCED VIBRATIONS

Objects such as a steel rod, tuning fork or a glass can be made to vibrate by striking them.
i.e. A single force is applied for a short time.

These objects will vibrate ***at their natural frequency*** (which depends on their physical dimensions and the material of which they are made.

However, if they are made to vibrate by an application of a periodic driving force (i.e. one applied with a regular frequency) with the same frequency as that of the driving force, this is called ***forced vibration***.

Example

A tuning fork that is struck and held in the hand is very difficult to hear unless it is placed close to the ear.

If it is struck and then placed in contact with a table, it is heard very clearly.

The table is forced to vibrate at the same frequency as the tuning fork. A louder sound is produced because a much larger mass of air in contact with the table is made to vibrate for a shorter period of time.

In musical instruments such as violins and guitars, the wooden frame and the enclosed air columns are forced to vibrate with the same frequency as the strings, thereby increasing the sound intensity.

RESONANCE

When the frequency of the periodic driving force coincides with the natural frequency of the body being forced to vibrate, a large amplitude of vibration is developed.

This phenomenon is called ***resonance***.

Examples

1. A child's swing. Small pushes given when the swing is about to move back from its maximum displacement will greatly increase the amplitude of the swing.

2. Stretched strings or air columns have a natural frequency of vibration. Sound waves of the same frequency produced by a vibrating source will cause them to resonate.

3. Parts on a car will vibrate or rattle at particular speeds. The vibrations from the tyres on the road match the natural frequency of the parts and cause them to vibrate significantly.

4. Dipping a finger in wine and rubbing it gently around the top of a wine glass will cause it to "ring". The ridges of the fingerprint cause small vibrations that match exactly the natural frequency of the air column inside the glass so it resonates.

BEATS

Beats refer to the unusual waxing and waning of sound that is produced when two sounds of similar amplitude but slightly differing frequencies occur at the same time.

i.e. ***The sound rises and falls in a regular pattern*.**

Beats are another example of interference between sound waves. The two waves superimpose and produce a single wave whose amplitude varies with time.

The resultant single wave is generated by the algebraic sum of the displacements of the individual waves.

PHYSICS IN ACTION

Effects of superposition and interference

Beats

If two sound waves of equal amplitude and slightly different frequency are added together, they superpose to form a resulting sound with a regular pulsation, known as a beat, as shown in Figure 10.4.6. At time X in the figure, the waves are in phase, producing a maximum in the sound; then at time Y, the waves are out of phase, producing a minimum in the sound. The beat frequency is given by $f_{beat} = |f_2 - f_1|$. This phenomenon can be used to tune a musical instrument against a standard tone. It is also the reason why twin-engine aircraft often have an oscillating drone, and is due to the slightly different speeds of the two engines.

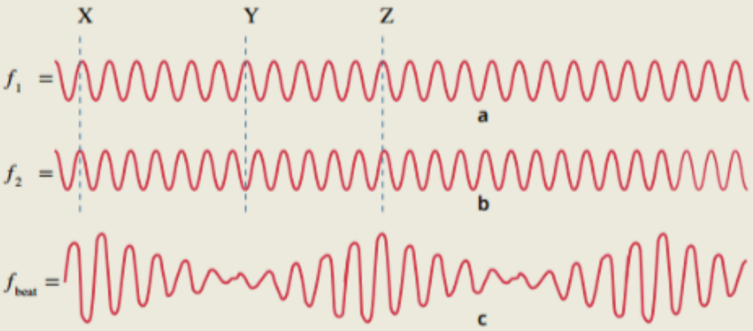


FIGURE 10.4.6 Superposition of two waves of slightly different frequencies f_1 and f_2 give rise to a wave with an oscillating amplitude.

Noise-cancelling headphones

Noise-cancelling headphones record the outside unwanted noise, and then produce a wave that is exactly opposite in phase, i.e. where there is a maximum in the amplitude of the recorded noise, there will be a minimum in the amplitude of the sound produced by the headphones. The superposition of the unwanted noise with the 'anti-noise' wave occurs over a time faster than your brain can process and thus creates destructive interference and effectively allows you to hear only the sound from your phone or device.

The frequency at which beats are produced is given by:

$$\boxed{f_{beat} = |f_2 - f_1|}$$

where f_{beat} = the beat frequency of the resultant wave (Hz)
f_1, f_2 = the frequency of the two waves generated at the same time (Hz)

Example

When a piano note and a tuning fork of frequency 360 Hz are sounded together, 6.5 beats per second are heard. One prong of the tuning fork is loaded with a small piece of wax and the number of beats heard per second is increased. What is the frequency of the piano note?

Putting wax on the fork will slow down the tines so its frequency is lowered.

Since this increases the beat frequency, then the piano note must be higher than the fork.

$$\therefore \quad f_{beat} = f_{piano} - f_{fork}$$
$$\Rightarrow \quad f_{piano} = 6.5 + 360$$
$$= 366.5 \text{ Hz}$$

$\therefore$ The piano note is 366.5 Hz.

INTENSITY OF SOUND

Sound intensity: the rate of flow of sound energy through a unit area perpendicular to the direction of travel of the sound wave at a given point.

$$\boxed{I = \frac{E}{At} = \frac{P}{A}}$$

where I = intensity of sound (Wm^{-2})
P = the power of the generated sound (W)
$= {}^{E}/_{t}$

The intensity of the sound is determined by its ***amplitude*** - the greater the amplitude, the greater the intensity.

Note: Loudness and intensity are related but are ***not*** the same.

Loudness can vary for different listeners as it is a sensation in the consciousness, and it varies for different frequencies as the human ear is not equally sensitive to all frequencies.

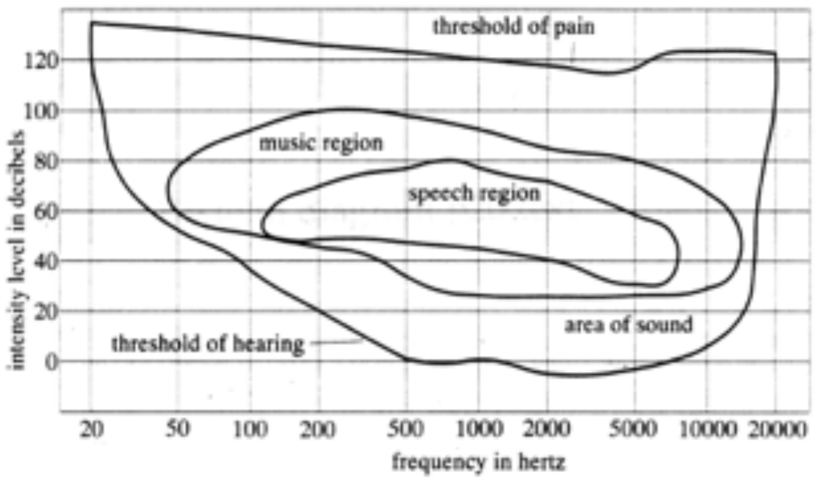
i.e. A high frequency sound may not seem as loud as a low frequency sound of the same intensity.

Sound Intensity Level

The relationship between loudness and intensity is a ***logarithmic*** one.

The human ear can detect sounds with an intensity as low as 1.00×10^{-12} Wm^{-2} (the threshold of hearing) and as high as 1.00 Wm^{-2} (the threshold of pain).

A sound intensity of 1.00×10^{-12} Wm^{-2} at 3000 Hz is called the ***reference sound***. This frequency is the one to which the ear is most sensitive.



The intensity level of a sound is measured in decibels (dB) and is calculated using:

$$\boxed{L = 10\log\frac{I}{I_o}}$$

where L = sound intensity level (dB)
I = the intensity of the sound (Wm^{-2})
I_o = reference sound level
= 1.00×10^{-12} Wm^{-2}

The decibel scale runs from 0 (threshold of hearing) to 120 dB (threshold of pain) and beyond.

EXTENSION

The decibel scale

The human ear can detect sound intensities from 10^{-12} W m^{-2} to 1 W m^{-2}. Therefore, a sound intensity of 10^{-11} W m^{-2} is 10 times larger than an intensity of 10^{-12} W m^{-2}. For everyday use sound levels are usually measured in decibels, which is a logarithmic scale where:

$$I\,(\text{dB}) = 10\log\left(\frac{I}{I_0}\right)$$

where I_0 is the threshold intensity of hearing (10^{-12} W m^{-2})
I is the sound intensity.

Some typical values for sound intensity are given in Table 10.6.1. When viewing the table, bear in mind every 10 dB means 10 times the sound intensity.

The long-term effect on a person's hearing also depends on how long a person is exposed to loud noise. It is an occupational health and safety requirement that people who regularly work in a loud environment wear ear protection.

TABLE 10.6.1 Examples of common sounds, their sound intensity in W m^{-2}, the relative sound intensity compared to the threshold of hearing ($I_0 = 10^{-12}$), and the equivalent sound intensity in dB.

Sound	Sound intensity, I (W m^{-2})	I/I_0	I (dB)
hearing threshold (I_0)	10^{-12}	$10^0 = 1$	0
whisper	10^{-10}	10^2	20
normal conversation	10^{-6}	10^6	60
heavy traffic	10^{-5}	10^7	80
loud truck close by	10^{-3}	10^9	90
fighter jet at 300 m	10^{-2}	10^{10}	100
front row of a rock concert; petrol-powered tools	10^{-1}	10^{11}	110
threshold of pain	10^1	10^{13}	130

Examples

1. At what sound intensity does a human experience pain rather than sound?

This occurs at 120 dB.

$$L = 10\log[I/I_o]$$
$$\Rightarrow \quad 120 = 10\log[I/1.00\times10^{-12}]$$
$$\Rightarrow \quad 12 = \log[I/1.00\times10^{-12}]$$
$$\Rightarrow \quad 1.00\times10^{12} = \frac{I}{1.00\times10^{-12}}$$
$$\Rightarrow \quad I = 1.00\ Wm^{-2}$$

$\therefore$ The sound intensity $= 1.00\ Wm^{-2}$.

2. During a check on a printing press working at a newspaper, a technician notes the sound intensity level at 95.0 dB at 3.00 m from the machine. A nearby machine then starts up and the ***sound intensity doubles***. What is the new sound intensity level?

ORIGINAL LEVEL $\quad L = 10\log[I_1/I_o]$

$\Rightarrow \quad 95.0 = 10\log[I_1/1.00\times10^{-12}]$ **Eqn 1**

NEW LEVEL $\quad L = 10\log[2I_1/1.00\times10^{-12}]$ (intensity doubles)

$= 10\log 2 + 10\log[I_1/1.00\times10^{-12}]$

$= 3.01 + 95.0$ (from Eqn 1)

$= 98.01$ dB

$\therefore$ New sound intensity level = 98.0 dB.

Note: Doubling the sound intensity only increases the sound intensity level by 3.00 dB.

Effect Of Distance On Sound Intensity

Sound intensity decreases by the square of the distance from the source.

$$\boxed{I \propto \frac{1}{d^2}}$$

This explains why a loud sound close to the ear is very bad, but the same sound further away does not cause nearly as much damage.

To compare two intensities of a sound at different distances:

$$\boxed{\frac{I_1}{I_2} = [\frac{d_2}{d_1}]^2}$$

where I_1 = intensity at a distance d_1 from the source (Wm^{-2}).
I_2 = intensity at a distance d_2 from the source (Wm^{-2}).
d_1, d_2 = distance from the source (m).

PHYSICSFILE

Inverse square laws

The law for the intensity of light around a point source of light is an example of an inverse square law. There are a number of important inverse square laws in physics.

Some others you will encounter in physics are Coulomb's law for the force between electric charges, and Newton's law of universal gravitation, for the gravitational force between any two masses.

In all these cases, something can be imagined to be spreading out evenly from a point. The intensity of this 'something' at a certain distance will therefore be inversely proportional to the area of the sphere over which it is spread. As the formula for the surface area of a sphere is $A = 4\pi r^2$, the intensity will therefore decrease with the square of the distance.

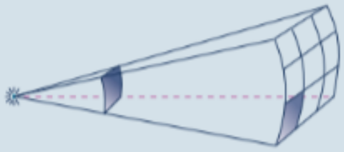


FIGURE 10.6.3 The inverse square law for a light source. At a distance of 1 m a square is drawn perpendicular to the rays. At a distance of 3 m from a point source, the light will be spread over an area nine times as large as that at 1 m. The light will therefore appear only one-ninth as bright.

Examples

1.

INTENSITY AND DISTANCE 1

Mia hears a siren sounding 10 m away. By what factor would the intensity of the sound from the siren change if Mia was to move to a distance of 50 m from the siren?	
Thinking	**Working**
Intensity, I, will decrease with the square of the distance, r, from the source. The ratio of the intensity at 50 m to the original intensity at 10 m is the factor required. Identify the initial values I_0 and r_0 and the final values I_f and r_f.	$r_0 = 10$ m $I_0 \propto \frac{1}{r_0^2}$ then $I_0 \propto \frac{1}{10^2}$ $r_f = 50$ m $I_f \propto \frac{1}{r_f^2}$ then $I_f \propto \frac{1}{50^2}$
Determine the relationship between the variables.	$\frac{I_f}{I_0} = \frac{\frac{1}{r_f^2}}{\frac{1}{r_0^2}}$ $\frac{I_f}{I_0} = \frac{r_0^2}{r_f^2}$
Evaluate.	$\frac{I_f}{I_0} = \frac{10^2}{50^2}$ $\frac{I_f}{I_0} = 0.04$

2.

INTENSITY AND DISTANCE 2

An ambulance started its siren 200 m after leaving the scene of an accident. After a short time the intensity of the sound was measured as being a quarter of the original. Assuming the volume of the siren hadn't changed, how far away was the ambulance when the intensity was measured?	
Thinking	**Working**
Intensity, I, will decrease with the square of the distance, r, from the source. The expression $\frac{I_f}{I_0} = \frac{r_0^2}{r_f^2}$ can be used. Identify the variables r_0 and $\frac{I_f}{I_0}$.	$r_0 = 200\,\text{m}$ $\frac{I_f}{I_0} = \frac{1}{4}$
Rearrange the expression and evaluate.	$\frac{I_f}{I_0} = \frac{r_0^2}{r_f^2}$ $r_f^2 = \frac{r_0^2 I_0}{I_f}$ $r_f^2 = \frac{200^2}{0.25} = 200^2 \times 4 = 160\,000$ $r_f = 400\,\text{m}$

3. A sound is measured at an intensity level of 90.0 dB at a distance of 3.00 m. What would be the new sound intensity level at a distance of 12.0 m from the source?

To calculate I_1 (at 3.00 m):

$$L = 10 \log [I_1 / I_o]$$

$$\Rightarrow \quad 90.0 = 10 \log [I_1 / 1.00 \times 10^{-12}]$$

$$\Rightarrow \quad 1.00 \times 10^{9} = \frac{I_1}{1.00 \times 10^{-12}}$$

$$\Rightarrow \quad I_1 = 1.00 \times 10^{-3}\ \text{Wm}^{-2}$$

To calculate I_2:

$$I_1 / I_2 = (d_2 / d_1)^2$$

$$\Rightarrow \quad I_2 = I_1 (d_1)^2 / (d_2)^2 = \frac{(1.00 \times 10^{-3})(3.00)^2}{(12.0)^2} = 6.25 \times 10^{-5}\ \text{Wm}^{-2}$$

To calculate the new level:

$$L = 10 \log [I_2 / I_o] = 10 \log [6.25 \times 10^{-5} / 1.00 \times 10^{-12}] = 77.96\ \text{dB}$$

$\therefore$ New sound intensity level = 78.0 dB.

4. A speaker that is rated at 50.0 W is placed against a wall in a large room. It is engineered to produce hemispherical wavefronts forward of the speaker, with the speaker acting as a point source.

Assume that the speaker is only 1.00 % efficient.

i.e. Only 1.00 % of the electrical energy input is converted to sound energy. (This is true of most speakers.)

(a) Calculate the power generated as sound at the speaker.
(b) Determine the surface area of a wave front 5.00 m from the speaker.
(c) What is the intensity of the sound wave at 5.00 m?
(d) What sound level would be measured by a sound meter at this point?

(a)

$$P_{sound} = \frac{1.00}{100} \times 50.0$$
$$= 0.500 \text{ W}$$

∴ Power generated as sound = 0.500 W.

(b)

$$\text{Surface area} = \text{area of a hemisphere}$$
$$= 2\pi r^2$$
$$= 2\pi(5.00)^2$$
$$= 1.571 \times 10^2 \text{ m}$$

∴ Surface area = 1.57×10^2 m.

(c)

$$I = \frac{P}{A}$$
$$= \frac{0.500}{1.571 \times 10^2}$$
$$= 3.183 \times 10^{-3} \text{ Wm}^{-2}$$

∴ Intensity = 3.18×10^{-3} Wm^{-2}.

(d)

$$L = 10 \log [I / I_o]$$
$$= 10 \log [3.183 \times 10^{-3} / 1.00 \times 10^{-12}]$$
$$= 95.03 \text{ dB}$$

∴ Sound intensity level = 95.0 dB.